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BUBBLEVERSE

Main Book

Building a Model of the Universe from Observation

Chapters 1–6

Glossary and references

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Contents

Preface..... 3

Chapter 1 What We Actually Observe.....5

Chapter 2 Measuring the Present Expansion Rate.....13

Chapter 3 The Hubble Tensi.....18

Chapter 4 Testing Early-Universe Solutions.....23

Chapter 5 What the Model Explains and What Remains Unknown.....31

Chapter 6 Experimental and Computational Record.....34

Glossary.....49

References..... 54

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0.42

Preface

Cosmology is often presented as a sequence of established facts: the Universe expands, distant galaxies recede, the cosmic microwave background records an earlier thermal state, and a small set of parameters describes the large-scale evolution of the cosmos. That presentation is useful, but it can hide a more fundamental scientific question: how much of this picture follows directly from observation, how much depends on inference, and where do the assumptions enter?

This book approaches cosmology by rebuilding the argument from the evidence outward. The aim is not to begin with a preferred model and search for observations that support it. Instead, the investigation asks what can actually be established, which conclusions depend on a particular theoretical framework, which measurements are genuinely independent, and which uncertainties remain capable of changing the interpretation.

This distinction matters because cosmological observations rarely measure the quantities of greatest theoretical interest directly. A redshift is observed. An angle is observed. A flux is observed. A spectrum is observed. A telescope records photons. A spectrograph records wavelengths. An interferometer records changes in optical path length. From these signals, distances, expansion rates, densities, ages, masses, and model parameters are reconstructed through layers of calibration, physical modelling, and statistical inference. Beyond those reconstructions lie still deeper interpretations: dark matter, dark energy, the hot early Universe, inflation, and the cosmological framework used to connect them.

Precision does not make every quantity equally direct.

A scientifically useful reconstruction of the Universe must therefore preserve the boundary between what an instrument measures and what a model allows us to conclude from that measurement. A successful model may provide an extraordinarily accurate description of observations without automatically becoming the final physical explanation. Conversely, the absence of a directly identified microscopic mechanism does not erase the observations that motivated the model.

The investigation begins with the empirical ground beneath those questions. It first asks what is actually observed, then moves outward toward reconstructed expansion, disagreements among inference chains, proposed explanations, and the computational tests used to examine them.

The working rule throughout the book is simple:

Start with the signal. Preserve the measurement. Make every inferential step visible.

How to Read This Book

This book is organized in layers.

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0.42

The main chapters tell the scientific story. They are intended to be readable without professional training in cosmology, statistics, or scientific computing. Technical ideas are introduced when they become necessary, but the reader is not expected to understand the internal Bubbleverse research machinery in order to follow the scientific argument.

The Glossary provides short definitions of the recurring scientific, statistical, computational, and Bubbleverse-specific terms used throughout the book.

The technical appendices preserve the deeper mathematical, numerical, computational, and provenance material behind the main narrative. They are intended for readers who want to inspect the machinery rather than merely follow the scientific story.

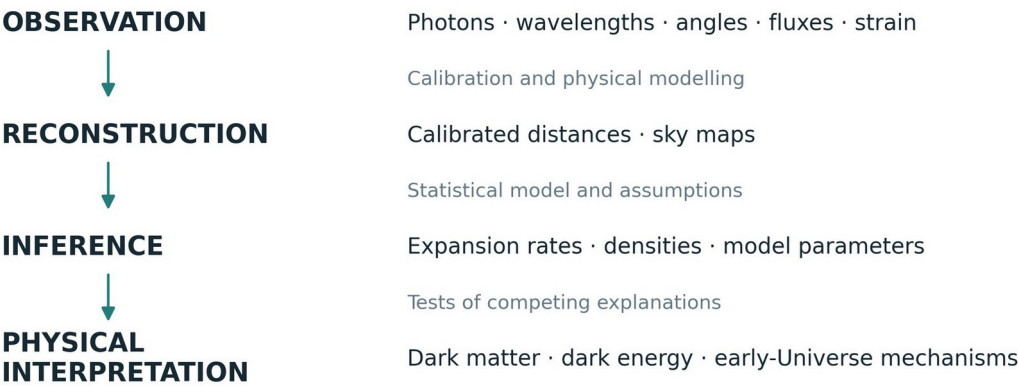
The Bubbleverse-Model Repository appendix documents how the accepted scientific state is stored, versioned, tested, promoted, and traced inside the dedicated Bubbleverse model repository.

The final appendix contains the Bubbleverse Q-Journals. These journals preserve the chronological research record behind the book: what was asked, what was tested, what failed, what changed, and why the investigation moved to the next question.

The guiding distinction throughout remains the same: observation, reconstruction, inference, and physical interpretation are connected, but they are not the same thing.

Figure 1. The evidential layers used throughout the book. Calibration and physical modelling connect observed signals to reconstructed quantities, parameter inference and physical interpretation. The layers are connected, but they are not interchangeable. Schematic based on the Preface and Chapter 1.

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Chapter 1 — What We Actually Observe

The Observable Universe

Astronomy does not provide a view of the universe from outside. Every empirical statement begins inside one causally accessible region: the observable universe.

Light and other signals require time to travel. There is therefore a limit to the region from which information can have reached us during cosmic history. Observations can constrain the contents, geometry, dynamics, and past evolution of this observable region, but they do not directly determine the size or topology of the universe as a whole.

This distinction rules out several claims that observation alone cannot establish. We do not directly know whether the entire universe is finite or infinite. We do not observe a physical outer edge. We do not observe a spatial centre from which cosmic expansion proceeds. Nor can observations inside our causal region determine everything that may exist beyond it.

The universe and the observable universe must therefore not be treated as synonyms.

Within the region we can observe, however, the evidence is extraordinarily rich.

Matter is distributed hierarchically across an enormous range of scales. We observe planets, stars, stellar remnants, gas, dust, galaxies, galaxy groups, galaxy clusters, filaments, and vast underdense regions. The distribution is strongly inhomogeneous on small and intermediate scales, while sufficiently large volumes approach statistical homogeneity and isotropy.

This large-scale structure is not a theoretical abstraction. Modern spectroscopic surveys measure the positions and redshifts of millions of galaxies and quasars, allowing the three-dimensional distribution of cosmic structure to be reconstructed statistically. The second major data release of the Dark Energy Spectroscopic Instrument, for example, uses more than fourteen million galaxies and quasars in its baryon acoustic oscillation analysis.

The universe is therefore not observed as a smooth static background. It is observed as structured, evolving, and dynamic.

Looking Backward in Time

Because light travels at a finite speed, distance also provides access to earlier cosmic epochs.

A nearby galaxy is seen relatively recently in its history. A very distant galaxy is observed as it existed billions of years ago. Deep observations therefore function as a form of cosmic archaeology: different distances expose different stages of the universe's development.

This does not mean that astronomers literally watch a single galaxy evolve over billions of years. Instead, populations observed at different redshifts reveal systematic changes in properties such as star-formation activity, galaxy populations, chemical enrichment, quasar activity, clustering, and the abundance of massive structures.

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The observation of systematic evolution with redshift is itself empirical. Translating a particular redshift into an exact cosmic age requires a cosmological model.

That distinction will recur frequently: the signal can be direct even when the conversion into a physical historical quantity is model dependent.

The Cosmic Microwave Background

One of the most important signals in all of cosmology is present in every direction in the sky.

The cosmic microwave background, or CMB, is an almost isotropic field of microwave radiation whose spectrum is extraordinarily close to that of a thermal blackbody. A precision determination of its mean thermodynamic temperature gives

T C M B=2.72548±0.00057 K .

The temperature is not perfectly uniform. Tiny direction-dependent fluctuations are observed, together with a characteristic pattern of polarization.

These fluctuations are usually analysed statistically through angular power spectra.

Modern CMB experiments measure temperature correlations and polarization combinations conventionally denoted TT, TE, and EE. Their spectra contain a sequence of prominent acoustic peaks.

These peaks are observations in the statistical structure of the microwave sky.

Their physical interpretation is a separate step.

Within the standard hot-universe picture, the pattern is understood as the imprint of acoustic oscillations in a tightly coupled photon–baryon plasma before photons were able to propagate freely through the universe. This interpretation is exceptionally successful, but the distinction remains important: detectors observe microwave photons, anisotropies, polarization, and angular correlations. The early plasma is the physical explanation constructed from those observations.

Planck measured this structure with high precision, while the Atacama Cosmology Telescope provides an independent high-resolution measurement of the CMB temperature and polarization spectra.

The CMB therefore provides both a direct observed radiation field and one of the most powerful model-dependent windows onto the physical conditions of the early universe.

It is not, however, a direct observation of a singularity at $t=0$.

Redshift and the Evidence for Cosmic Expansion

The statement that the Universe expands sounds deceptively simple. It is tempting to imagine galaxies flying outward through pre-existing space from some central explosion.

That picture is not what relativistic cosmology means by expansion.

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0.42

Spectral lines from distant galaxies are systematically shifted toward longer wavelengths.

The dimensionless redshift is defined by

$$Z = \lambda_{\text{obs}} - \lambda_{\text{emit}}$$

,

λ_{emit}

Where λ_{emit} is the emitted wavelength and λ_{obs} is the wavelength measured by the observer.

At relatively low cosmological redshift, distance and recession behaviour follow an approximately linear relation associated with the Hubble law. At greater distances the relation becomes more complex, and its interpretation requires an explicit description of cosmic geometry and expansion history.

In a homogeneous and isotropic description of the Universe, expansion is represented by a time-dependent scale factor, $a(t)$. With the conventional normalization $a(t_0)=1$ at the present epoch, cosmological redshift is related to the scale factor by

$$1+z = \frac{a(t_0)}{a(t_{\text{em}})}$$

$$= \frac{1}{a(t_{\text{em}})}$$

.

$a(t_{\text{em}})$

The normalization itself is conventional; the physical information lies in ratios of the scale factor between epochs.

The corresponding expansion rate is

$$H(t) = \frac{\dot{a}(t)}{a(t)}$$

$a(t)$.

A positive value of H corresponds to expansion, while a negative value would correspond to contraction. Establishing that $H(z) > 0$ over an observed redshift interval therefore tests whether the reconstructed cosmological geometry is expanding throughout that interval. The scientific question is consequently more precise than asking whether distant galaxies are moving away from us. We must ask whether observations made at different distances and cosmic epochs are consistent with a common evolving scale factor.

It is useful to keep several quantities separate: observed wavelength, redshift, distance, recession velocity, the cosmic scale factor, $H(z)$, and the present expansion parameter H_0 . They are related, but they are not identical measurements.

The crucial distinction is between observation and inference. Redshifts, fluxes, angular separations, spectral features, and time intervals are observational quantities. A reconstructed function such as $H(z)$, by contrast, is generally inferred from those observations under specified assumptions.

The evidence for expansion is therefore stronger than the existence of cosmological redshifts alone. The relevant question is whether physically different observations of redshift, distance, and time can be described consistently by the same evolving cosmic geometry.

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This separation between measurement and geometry is not semantic nitpicking. Later questions about the Hubble constant and dark energy depend critically on knowing where the measurements end and the cosmological assumptions begin.

Supernovae and the Discovery of Late-Time Acceleration

Type Ia supernovae provide one of the clearest examples of the distinction between an observed relation and the physical conclusion drawn from it.

Their light curves can be standardized sufficiently well to compare their relative luminosity distances across cosmological redshift.

The original high-redshift supernova programmes led by Riess and collaborators and by Perlmutter and collaborators independently found that distant Type Ia supernovae were inconsistent with the distance–redshift relation expected in simple matter-dominated, non-accelerating cosmologies.

Modern samples are vastly larger. Pantheon+ contains 1,701 light curves corresponding to 1,550 distinct Type Ia supernovae over approximately $0.001 < z < 2.26$.

The primary observation is the standardized supernova distance–redshift relation.

Within Friedmann–Lemaître–Robertson–Walker cosmology and general relativity, that relation strongly supports a period of accelerated late-time cosmic expansion.

The acceleration is therefore a powerful physical inference, but it does not by itself identify its cause.

The term dark energy is used for the component, field, modification, or effective phenomenon responsible for reproducing this accelerated behaviour in a cosmological model. The name does not imply that a dark-energy particle or field has been directly isolated experimentally.

The observations establish the expansion history far more securely than they establish the microscopic explanation for it.

Supernovae constrain the integrated expansion history rather than measuring $H(z)$ at a single redshift. This distinction is important. A luminosity-distance relation can provide powerful evidence for an evolving cosmic geometry, but recovering a local expansion rate from it requires additional inference. The observed distance–redshift relation and the reconstructed expansion history should therefore not be treated as identical quantities.

Baryon Acoustic Oscillations

The large-scale distribution of matter contains another striking statistical feature: a preferred clustering scale known as the baryon acoustic oscillation, or BAO, scale.

The feature is measured in the spatial correlations of galaxies and quasars and can also be extracted from the Lyman- α forest.

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DESI has measured this structure across several redshift ranges with enormous spectroscopic samples.

The clustering feature itself is observed.

Its use as a cosmic standard ruler introduces additional physics.

In the standard interpretation, the scale is the late-time remnant of acoustic propagation in the photon–baryon plasma of the early universe. The physical size of that ruler is commonly related to the sound horizon r_d .

The radial BAO measurement is especially valuable for reconstructing expansion because the observed redshift separation of the acoustic feature is related to the combination $H(z)r_d$. BAO therefore does not determine $H(z)$ in complete isolation. Converting the observed ruler into an absolute expansion rate requires either an external calibration of r_d or assumptions about the early-universe physics that determines it.

This dependence does not remove BAO as evidence for an evolving distance–redshift geometry. Instead, it identifies precisely where early-universe physics enters the absolute calibration. That distinction will become especially important when different determinations of the cosmic expansion rate are compared.

Gravitational Lensing

Gravity does not merely affect the motion of matter. It also changes the propagation of light.

Foreground mass can distort, magnify, multiply image, or statistically shear radiation emitted by more distant objects. This phenomenon, gravitational lensing, is observed across a wide range of scales.

Lensing can therefore be used to reconstruct information about the integrated gravitating mass along the line of sight.

The CMB itself is gravitationally lensed by intervening cosmic structure. Planck reconstructed this lensing field with high statistical significance, providing a direct statistical probe of the matter distribution between the surface of last scattering and the observer.

The signal is particularly valuable because it depends on gravitating matter rather than merely on emitted light.

The Missing-Mass Phenomenon

One of the most persistent empirical facts in modern astrophysics is that the gravitational behaviour of many systems cannot be accounted for by the observed luminous baryonic material alone within conventional gravitational modelling.

Galaxy rotation curves provide one major example.

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In the SPARC galaxy sample, measurements of rotationally supported galaxies reveal a tight relation between the total centripetal acceleration inferred from the motion of matter and the acceleration expected from the observed baryonic distribution. At low accelerations the two cease to agree in the way a baryon-only Newtonian model would predict.

Galaxy clusters provide a physically different test.

In systems such as the Bullet Cluster, the dominant X-ray-emitting baryonic plasma can be spatially displaced from the dominant gravitating mass reconstructed from lensing. This creates a separation between where most of the ordinary gaseous matter is observed and where the gravitational mass is inferred to lie.

Together with CMB and large-scale-structure constraints, these observations motivate the standard conclusion that the universe contains a substantial non-luminous gravitating component.

The safest empirical statement is nevertheless more specific than the everyday phrase “dark matter has been observed.”

What has been observed are gravitational effects that require either additional gravitating mass or a successful alternative description of gravity capable of reproducing the same body of phenomena.

A predominantly non-baryonic dark-matter component is currently the most successful common explanation across the full cosmological framework.

Its microscopic identity remains unknown.

A WIMP, axion, ultralight field, sterile neutrino, or other proposed dark-matter candidate should therefore not be promoted to observational fact until independent evidence identifies it.

The gravitational phenomenon is robust. The particle identity is not.

Gravitational Waves

Cosmology and astrophysics no longer rely exclusively on electromagnetic radiation.

In 2015 the LIGO detectors recorded GW150914, a transient strain signal detected by two interferometers and consistent with the merger of two compact objects described as stellar-mass black holes within general relativity.

This was the first direct detection of gravitational waves.

GW170817 later provided a gravitational-wave signal from a binary neutron-star inspiral together with electromagnetic counterparts. Its network signal-to-noise ratio was approximately 32.4.

These events opened a genuinely different observational channel.

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The detectors do not observe an image of spacetime. They record correlated changes in interferometer arm lengths whose temporal structure is consistent with propagating gravitational waves.

The interpretation of the sources relies on relativistic waveform modelling, but the existence of gravitational-wave signals is now an observational fact.

Modern astronomy is consequently a multi-messenger discipline. Information reaches us through electromagnetic radiation, gravitational waves, neutrinos, and high-energy particles.

Independent channels matter because conclusions supported by different physics are harder to dismiss as artefacts of a single observational method.

Chemical Evidence from the Early Universe

Spectroscopy also reveals the chemical composition of astronomical matter.

Hydrogen, helium, deuterium, and heavier elements leave identifiable absorption and emission features. Particularly metal-poor absorption systems can be used to estimate primordial light-element abundances.

A representative precision determination of primordial deuterium gives

$$(D/H)_P = (2.527 \pm 0.030) \times 10^{-5}$$

–5.

This number is reconstructed from observed absorption spectra.

Its interpretation as a primordial abundance and its connection to Big Bang nucleosynthesis require additional physical modelling.

Within standard nuclear and cosmological physics, light-element abundances provide a strikingly successful test of a universe that once existed in a much hotter and denser state. The inference reaches much further back in cosmic history than direct electromagnetic imaging, but it is correspondingly less direct.

This distinction is useful: observation can preserve information about an epoch without giving us a photograph of that epoch.

A Hotter and Denser Past

Several physically different observations converge on the conclusion that the universe was once much hotter and denser than it is today.

The most important lines include cosmic expansion, the cosmic microwave background, and primordial light-element abundances.

The power of this conclusion does not arise from any one observation in isolation.

Expansion implies that matter and radiation occupied a denser configuration in the past.

The thermal microwave background supplies an observed relic radiation field. Nuclear abundances are consistent with reactions expected in a sufficiently hot early environment.

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These pieces form an unusually coherent empirical picture.

This strongly supports the hot Big Bang description of the observable universe's early evolution.

It does not demonstrate a physical singularity.

A mathematical extrapolation toward $a=0$ within classical general relativity is not itself an observation of an initial point of infinite density. The observational evidence establishes a hot, dense early phase; what happened arbitrarily close to the beginning of the classical extrapolation remains a deeper theoretical question.

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Chapter 2 — Measuring the Present Expansion Rate

Cosmological Parameters Are Not Instrument Readings

Some of the most familiar numbers in cosmology are extraordinarily precise, but they are not all measured in the same sense.

Under the six-parameter flat Λ CDM model, cosmological parameters are inferred by fitting a physical model to measured cosmic microwave background spectra. Planck remains an important benchmark, with

–1 M pc

–1,

$H_0 = 67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$

$\Omega_m = 0.315 \pm 0.007$,

and

$\sigma_8 = 0.811 \pm 0.006$.

The crucial distinction is epistemic: no Planck detector directly measured H_0 as a local recession rate. The fundamental quantity is

$H(t) = \dot{a}(t)/a(t)$,

and the Hubble constant is its present value,

$H_0 \equiv H(t_0) = (\dot{a}/a)|_{t=t_0}$.

H_0 is therefore a cosmological parameter rather than a raw observable. Redshift, distance, gravitational-wave strain, time delays, and microwave-background anisotropies can all contribute to its determination, but none of them is itself H_0 .

This becomes especially important when comparing modern estimates. A combined analysis of Planck, SPT-3G, and ACT DR6 cosmic microwave background information gives, within flat Λ CDM,

–1 M pc

–1.

$H_0 = 67.24 \pm 0.35 \text{ km s}^{-1} \text{ Mpc}^{-1}$

This CMB-based value is obtained through an inference chain that connects the observed temperature and polarization spectra to acoustic and recombination physics and then to the modelled cosmic expansion history.

The late-universe route is physically different. The classical SHOES distance ladder uses geometric anchors, Cepheid variables, Type Ia supernovae, and Hubble-flow objects, yielding

–1 M pc

$H_0 = 73.04 \pm 1.04 \text{ km s}^{-1} \text{ Mpc}^{-1}$

–1.

More recent work combines multiple local distance indicators while explicitly accounting for shared calibrations and covariance. The H0 Distance Network reports the covariance-weighted baseline

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0.42

–1 M pc

$H_0 = 73.50 \pm 0.81 \text{ km s}^{-1} \text{ Mpc}^{-1}$

–1.

This does not constitute a new single measurement technique. It is a synthesis of several connected distance-calibration routes, with SHOES remaining an important part of the underlying provenance.

The local determination is closer to the late-time expansion than the CMB inference, but it is still not a direct reading of $\dot{a}/a$ at $z = 0$. Geometric anchors, stellar distance indicators, supernova calibration, finite-redshift Hubble-flow observations, peculiar-velocity corrections, and the distance–redshift relation all enter before H_0 is obtained.

The disagreement is therefore not equivalent to two identical instruments measuring the same raw quantity and returning incompatible readings. It is a disagreement between physically and statistically different inference chains that estimate the same cosmological parameter.

Independent Routes to the Present Expansion Rate

The distinction between measurement and inference becomes clearer when the principal routes to H_0 are placed side by side. There is no single epistemic category that adequately describes every “Hubble-constant measurement.”

The Local Distance Network

The local distance network begins with geometric distance anchors and uses calibrated stellar indicators, Type Ia supernovae or other Hubble-flow tracers, and finite-redshift distance–redshift information to infer the present expansion rate. Alternative stellar calibrators such as the tip of the red-giant branch (TRGB) and J-region asymptotic giant branch (JAGB) stars can test dependencies specific to Cepheid calibration, although two calibrators are not fully independent if they subsequently share anchors, supernova hosts, or Hubble-flow samples.

Alternative Stellar Calibrators

A published Chicago-Carnegie Hubble Program analysis reveals significant structure within the late-universe calibration evidence rather than a single alternative value of H_0 . Its JWST-only calibrations give

–1 M pc

$H_0 (\text{TRGB}) = 68.81 \pm 1.79 \text{ km s}^{-1} \text{ Mpc}^{-1} \text{ at } \pm 1.32 \text{ km s}^{-1} \text{ Mpc}^{-1}$

–1

and

–1 M pc

–1.

$H_0 (\text{JAGB}) = 67.80 \pm 2.17 \text{ km s}^{-1} \text{ Mpc}^{-1} \text{ at } \pm 1.64 \text{ km s}^{-1} \text{ Mpc}^{-1}$

A larger calibration sample combining HST and JWST TRGB measurements instead gives

–1 M pc

$H_0 = 70.39 \pm 1.22 \text{ km s}^{-1} \text{ Mpc}^{-1} \text{ at } \pm 1.33 \text{ km s}^{-1} \text{ Mpc}^{-1} \pm 0.70 \text{ km s}^{-1} \text{ Mpc}^{-1}$

–1.

The late-universe evidence therefore cannot be represented accurately by assigning the Chicago-Carnegie programme a single value near $68 \text{ km s}^{-1} \text{ Mpc}^{-1}$. Different calibration

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samples and pipelines produce meaningfully different central values. At the same time, these results are not fully independent of the broader local-distance network because geometric anchors, host galaxies, supernova information, and calibration structures can overlap. The differences are consequently evidence of internal methodological and calibration structure within late-universe H_0 inference rather than a separate, comparably precise anomaly.

Geometric Megamasers

Water megamasers provide a more geometrically distinct route. Positions, spectra, velocities, and accelerations in maser disks around supermassive black holes can be modelled to obtain galaxy distances without the conventional stellar distance ladder. The Megamaser Cosmology Project obtained

–1 M pc

–1.

$H_0=73.9\pm3.0$ k m s

At the relatively low redshifts of these systems, peculiar velocities remain an important limitation.

Strong-Lensing Time Delays

Strong gravitational lenses provide another independent physical route. Differences in the arrival times of multiple images of a variable background source, combined with lens imaging, stellar kinematics, line-of-sight structure, and a model of the lens mass distribution, constrain the time-delay distance.

TDCOSMO 2025 obtains

+3.9 k m s

–1 M pc

$H_0=71.6-3.3$

–1

For its primary analysis combined with a Pantheon+ constraint on Ω_m in flat Λ CDM.

The method avoids the Cepheid distance calibration and sound-horizon calibration, but its precision depends critically on gravitational-lens mass modelling, stellar kinematics, line-of-sight structure, and the treatment of the mass-sheet degeneracy.

Gravitational-Wave Standard Sirens

Gravitational-wave standard sirens offer a fundamentally different distance indicator. The waveform amplitude contains luminosity-distance information; combining that distance with a host-galaxy redshift or a statistical galaxy-redshift distribution allows H_0 to be inferred.

The original GW170817 standard-siren analysis gave

–1 M pc

+12.0 k m s

–1.

$H_0=70.0-8.0$

A later analysis combining the fourth Gravitational-Wave Transient Catalogue with the Dark Energy Survey Year 6 Gold galaxy catalogue obtained

+22.7 k m s

–1 M pc

$H_0=71.7-18.3$

–1

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From dark sirens. When combined with GW170817, the result becomes

–1 M pc

+11.9k m s

–1.

H 0=73.1–8.5

These constraints remain much broader than the leading CMB and local-distance determinations. They therefore do not yet decide the Hubble tension, but their different physical dependence makes them an important cross-check. The combined result must not be counted as statistically independent of the original GW170817 measurement because that event is reused in the combined analysis.

This route avoids Cepheid, TRGB, supernova absolute-luminosity, and sound-horizon calibration. It is not assumption-free, however. Waveform modelling, detector calibration, binary inclination, source populations, host identification, selection effects, and peculiar velocities remain relevant.

Current standard-siren constraints are still substantially broader than the most precise CMB and distance-network results, making them particularly valuable as an independent cross-check rather than, at present, the most precise adjudicator.

BAO and the Absolute Standard Ruler

Baryon acoustic oscillations provide yet another inference chain. BAO measurements primarily constrain quantities such as

$DM(z)/r_d$

And

$H(z)r_d$,

Where r_d is the sound horizon associated with the baryon-drag epoch.

BAO therefore does not provide an absolute H_0 without additional information fixing the physical ruler or otherwise breaking the relevant degeneracy.

DESI DR2 combined with a conservative Big Bang nucleosynthesis constraint on the baryon density gives, under flat Λ CDM,

–1 M pc

$H_0=68.51\pm0.58$ k m s

–1.

This result is independent of CMB anisotropy measurements in an important observational sense, but it is not model independent. The inference still relies on early-universe physics, sound-horizon calibration, and the specified cosmological framework.

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A Continuum of Inference

Figure 2. Principal inference routes to H_0 . The same cosmological parameter is reached through physically and statistically different chains involving geometric calibration, stellar indicators, gravitational lensing, gravitational waves, standard-ruler physics, and CMB modelling. Their different dependencies are central to the interpretation of agreement and disagreement among H_0 estimates.

These methods form a continuum of inference depth rather than a clean division into “direct” and “indirect” measurements.

The local distance network is dominated by geometric and astrophysical calibration.

Megamasers are more directly geometric but remain sensitive to velocity modelling.

Strong-lensing time delays combine geometry with gravitational modelling.

Standard sirens derive distance from gravitational-wave physics.

BAO requires a calibrated standard ruler.

CMB constraints infer H_0 through the most extensive global cosmological model chain.

Their scientific value lies precisely in these different dependencies. Agreement between genuinely independent routes is much more informative than agreement between several analyses that repeatedly use the same underlying observations, calibration, or model assumptions.

INFERENCE ROUTE	OBSERVATIONS AND DEPENDENCIES	H_0
Local distance network	Geometric anchors → stellar calibrators → SNe / Hubble-flow tracers → finite-redshift distance relation	→ 73.50 ± 0.81
Megamasers	Disk positions, velocities and accelerations → geometric galaxy distances + peculiar-velocity treatment	→ 73.9 ± 3.0
Strong-lensing time delays	Images + time delays + lens mass model + stellar kinematics + line-of-sight structure → time-delay distance	→ $71.6^{+3.9}_{-3.3}$
Gravitational-wave standard sirens	Waveform distance + host / statistical redshift + selection and inclination modelling	→ Currently broad
BAO + BBN	BAO geometry + calibrated sound horizon + baryon density + cosmological framework	→ 68.51 ± 0.58
CMB	TT / TE / EE spectra + acoustic physics + recombination physics + cosmological model (flat Λ CDM)	→ 67.24 ± 0.35

Shared local anchors, hosts and calibration samples can create covariance.
Values in $\text{km s}^{-1} \text{Mpc}^{-1}$. Distinct routes are not all statistically independent.

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Chapter 3 — The Hubble Tensi

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A Broader Conflict Than Planck versus SH0

The modern Hubble tension is broader than the traditional shorthand of “Planck versus **SHOES.**”

The strongest current benchmarks considered here include a combined CMB inference from Planck, SPT-3G, and ACT DR6,

$H_0(\text{CMB, flat } \Lambda\text{CDM}) = 67.24 \pm 0.35 \text{ km s}^{-1} \text{ Mpc}^{-1}$,

and the covariance-weighted local H0 Distance Network result,

-1 Mpc

$H_0(\text{local distance network}) = 73.50 \pm 0.81 \text{ km s}^{-1} \text{ Mpc}^{-1}$

-1.

Using the published marginal uncertainties in a simple Gaussian comparison, these constraints differ by approximately 7.1σ .

A Separate Low- H_0 Route

A standard-ruler route also lies on the lower- H_0 side.

DESI DR2 BAO combined with Big Bang nucleosynthesis gives, under flat ΛCDM ,

-1 Mpc

$H_0(\text{BAO+BBN}) = 68.51 \pm 0.58 \text{ km s}^{-1} \text{ Mpc}^{-1}$

-1.

The local-network baseline differs from this result by approximately 5σ .

This matters because the lower- H_0 region is not supported solely by the Planck CMB inference. DESI BAO combined with BBN reaches a similar region without using CMB anisotropy information.

It does not, however, provide a model-independent determination. Sound-horizon physics and the assumed cosmological framework remain part of the inference.

Different Numbers, Different Inference Chains

These results should not be interpreted as three identical measurements.

The CMB value is inferred through a global cosmological model and early-universe acoustic physics.

BAO+BBN avoids CMB anisotropy information but still depends on early-universe standard-ruler physics and the assumed cosmological framework.

The local distance network instead relies on geometric and astrophysical distance calibration, finite-redshift Hubble-flow information, peculiar-velocity treatment, and calibration covariance.

BUBBLEVERSE MAIN BOOK

0.42

They estimate the same parameter, but they do not arrive there through the same physical or statistical route.

Internal Structure in the Late-Universe Evidence

There is also important structure within the late-universe evidence itself. JWST-only TRGB and JAGB calibrations yield central values of 68.81 and $67.80 \text{ km s}^{-1} \text{ Mpc}^{-1}$, respectively, while a larger HST+JWST TRGB calibration sample gives $70.39 \text{ km s}^{-1} \text{ Mpc}^{-1}$. The covariance-weighted H0 Distance Network gives the still higher and substantially more precise value

-1 M pc

$-1.$

$H_0 = 73.50 \pm 0.81 \text{ km s}^{-1} \text{ Mpc}^{-1}$

These differences must remain visible rather than being compressed into a single category called “the local measurement.” They demonstrate genuine calibration and pipeline structure within late-universe H_0 inference. Their interpretation nevertheless requires care because overlapping anchors, host galaxies, supernova samples, and calibrations prevent the individual results from being treated as completely independent experiments.

An additional test removes Type Ia supernovae as the final rung of the conventional distance ladder. Using geometric anchors, Cepheids, direct host redshifts, explicit sample-selection modelling, and peculiar-velocity reconstruction, this two-rung analysis finds

-1 M pc

$-1.$

$H_0 = 71.1 \pm 1.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$

Under a simple Gaussian comparison, this remains approximately 2.7σ above the

-1 M pc

combined CMB value of $67.24 \pm 0.35 \text{ km s}^{-1} \text{ Mpc}^{-1}$

$-1.$ Because the analysis reuses Cepheid and

anchor information employed by the conventional local-distance programme, it is not a fully independent measurement. It nevertheless shows that removing the supernova rung does not by itself eliminate the high-versus-low H_0 discrepancy. At the same time, the lower central value demonstrates that sample selection, finite-redshift treatment, and peculiar-velocity reconstruction can materially shift the inferred local expansion rate. These effects therefore belong to the explanation problem, but they do not by themselves erase the tension.

The local evidence should therefore be treated neither as a perfectly homogeneous high- H_0 block nor as a collection of independent competing measurements. It is a correlated network with real internal structure, and that structure is itself part of the problem that any successful explanation must accommodate.

Independent Cross-Checks

Other physically distinct probes provide valuable cross-checks but still lack the precision required to decide the conflict alone. Megamasers give $H_0 = 73.9 \pm 3.0 \text{ km s}^{-1} \text{ Mpc}^{-1}$, while strong-lensing time delays give $H_0 = 71.6^{+3.9}_{-3.3} \text{ km s}^{-1} \text{ Mpc}^{-1}$ in the cited analysis. Current gravitational-wave standard-siren constraints are broader still. None presently has the combination of precision and independence required to adjudicate decisively between H_0 near 67 and the local-network value near $73.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$.

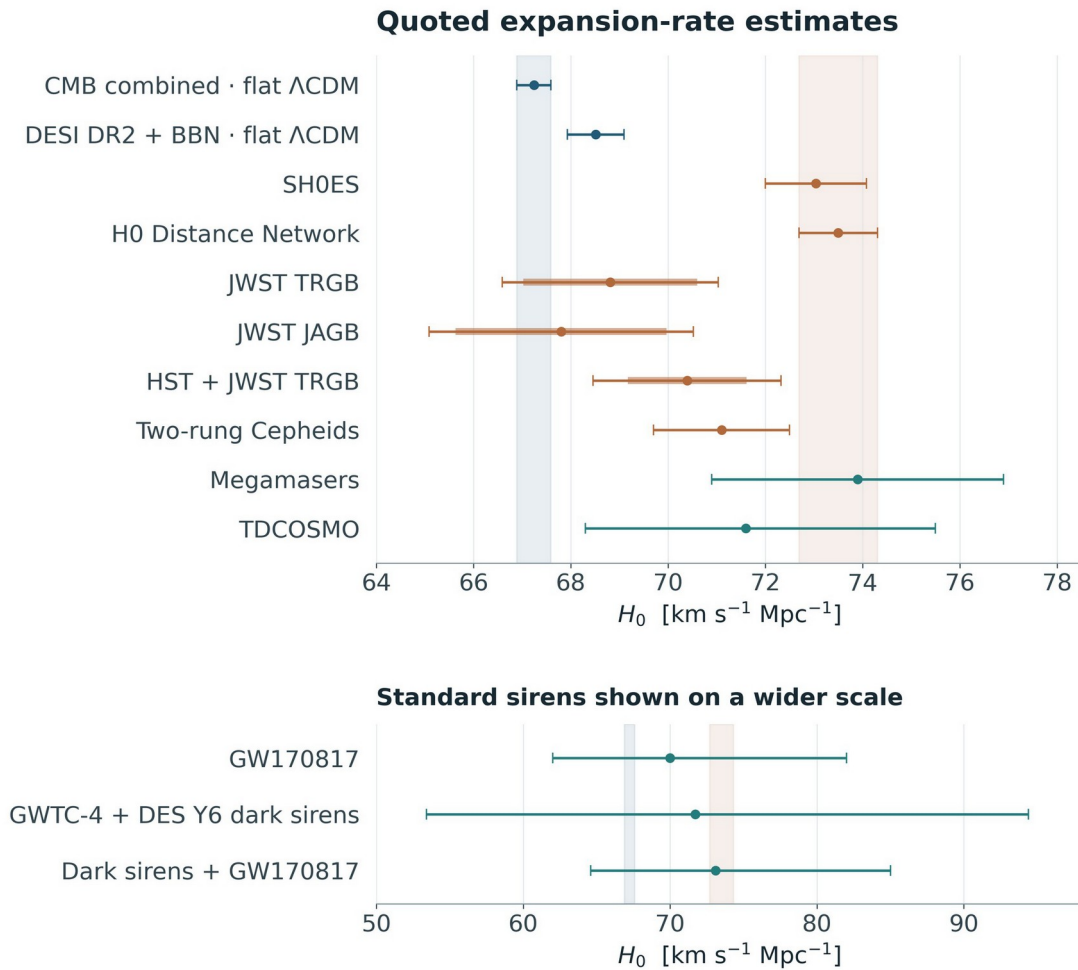
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0.42

Their importance lies in their different dependence on calibration and modelling assumptions. Megamasers avoid the conventional stellar distance ladder and sound-horizon calibration but remain sensitive to peculiar velocities. Strong-lensing time delays avoid those same central calibrations but depend on lens-mass modelling, stellar kinematics, line-of-sight structure, and the treatment of the mass-sheet degeneracy. Standard sirens derive luminosity distance from gravitational-wave information but retain waveform, detector, population, host, selection, and velocity dependencies.

The absence of a roughly one-percent H_0 determination that is simultaneously almost independent of both stellar distance-ladder calibration and early-universe sound-horizon calibration remains an important observational gap.

Figure 3. Expansion-rate estimates quoted in Chapters 2 and 3. Standard sirens use a wider horizontal scale. For the three TRGB and JAGB entries, thin bars combine the quoted uncertainty components in quadrature for display; thick bars show the statistical component. Other bars reproduce the quoted intervals. No cross-study covariance is modelled. GW170817 is reused in the combined siren result, and shared observations and calibrations prevent all entries from being treated as independent measurements.



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0.42

What the Tension Does—and Does Not—Establish

The combined evidence now supports a stronger conclusion than merely noting that different estimates of H_0 disagree. The discrepancy is a robust cosmological parameter tension across the inference structures examined here.

The covariance-weighted local H_0 Distance Network gives

-1 M pc

$H_0 = 73.50 \pm 0.81 \text{ km s}^{-1} \text{ Mpc}^{-1}$

-1 ,

whereas the combined Planck, SPT-3G, and ACT CMB analysis gives, under flat Λ CDM,

-1 M pc

-1 .

$H_0 = 67.24 \pm 0.35 \text{ km s}^{-1} \text{ Mpc}^{-1}$

The difference is $6.26 \text{ km s}^{-1} \text{ Mpc}^{-1}$. Combining the published marginal uncertainties in quadrature and neglecting cross-covariance between these principal chains gives a discrepancy of approximately 7.1σ .

The 7.1σ value is the comparison obtained from the published H_0 Distance Network baseline, $H_0 = 73.50 \pm 0.81 \text{ km s}^{-1} \text{ Mpc}^{-1}$, and the combined CMB inference quoted above. A subsequent 2026 reanalysis applying corrected Pantheon+ host-mass information while retaining the HODN covariance, redshifts, and velocities gives $H_0 = 73.264 \pm 0.806 \text{ km s}^{-1} \text{ Mpc}^{-1}$, corresponding to a reported 6.85σ difference from the same CMB benchmark.

Alternative corrected-Pantheon+ and DES-DOVEKIE implementations give similarly high values. These updates shift the precise numerical significance but do not remove the underlying high-versus-low H_0 discrepancy. The original HODN value should therefore be retained as the network consensus baseline rather than silently replaced by one subsequent supernova reanalysis.

A second low- H_0 route strengthens the conclusion. DESI DR2 BAO combined with Big Bang nucleosynthesis gives, under flat Λ CDM,

-1 M pc

$H_0 = 68.51 \pm 0.58 \text{ km s}^{-1} \text{ Mpc}^{-1}$

-1 ,

approximately 5σ below the local-network baseline under the same simple marginal

comparison. Because this route does not use CMB anisotropy likelihoods, the low- H_0 side cannot plausibly be reduced to a Planck-specific instrumental or analysis-pipeline effect.

The CMB and BAO+BBN inferences are not physically assumption-independent, however: both retain important dependence on standard early-universe and sound-horizon physics.

Several simple single-cause explanations can now be narrowed more sharply. JWST

Cepheid cross-checks do not reveal a crowding bias large enough to erase the high local expansion rate, and removing Type Ia supernovae as the final distance-ladder rung still

-1 M pc

leaves $H_0 = 71.1 \pm 1.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$

-1 . Ordinary peculiar velocities and standard cosmic

variance are also too small to provide the full required shift. Local calibration structure,

sample selection, finite-redshift treatment, and velocity modelling can therefore move the

inferred value and may contribute to the discrepancy, but no tested ordinary systematic

has been shown to reconcile the complete local, CMB, and BAO+BBN evidence by itself.

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0.42

The same standard must be applied to cosmological extensions. Several mechanisms can move the inferred value of H_0 upward, but that is not sufficient. A successful explanation must preserve the observed CMB acoustic structure, the BAO and Big Bang nucleosynthesis constraints, the internal structure of late-universe calibrations, and the physically distinct expansion-rate routes. Under this broader test, none of the remaining classes examined here is established as a robust complete solution. The approximately 7.1σ discrepancy remains a statement about conflicting parameter inferences; it is not a 7.1σ detection of new physics.

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0.42

Chapter 4 — Testing Early-Universe Solutions

What a Successful Solution Must Do

The Hubble tension cannot be solved merely by producing a model in which the inferred value of H_0 moves upward.

A successful early-Universe explanation must alter the calibration linking early-time physics to later cosmological distances while remaining compatible with the rest of the observations. The same model must still account for the measured cosmic microwave background, baryon acoustic oscillations, Big Bang nucleosynthesis constraints, and the other data used in the inference.

This is why reducing the sound horizon is necessary for many early-Universe proposals but is not sufficient. Changing one part of the cosmological ruler forces compensating changes elsewhere.

The Bubbleverse calculations therefore tested not only whether a mechanism could reach a higher H_0 , but what happened to the complete likelihood while it did so.

$n=3$ Early Dark Energy

Early dark energy, or EDE, provides one way of changing the pre-recombination expansion history. In the $n=3$ model examined here, a temporary additional energy component can reduce the effective early-Universe distance scale and allow higher inferred values of H_0 . A recent profile-likelihood analysis discussed in this investigation reports a region around $f_{\text{EDE}} = 0.09 \pm 0.03$ and $H_0 = 71.0 \pm 1.1 \text{ km s}^{-1} \text{ Mpc}^{-1}$. The Bubbleverse calculations likewise found higher- H_0 regions, but the numerical history showed why the existence of such a region is not enough to establish a solution.

The first difficulty was optimization. Early fixed- H_0 searches produced a non-monotonic profile and apparently large penalties at some points. Later multi-start and deterministic recovery showed that some of those results belonged to inferior numerical basins rather than to the best solutions available on the frozen likelihood surface.

That correction materially changed the interpretation. A poor optimizer result cannot be used as evidence against a physical model.

At the active $H_0 = 71.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ vector, the internally consistent serialized fixed-vector attribution gives

$\Delta \chi^2_{\text{total}}$

$2 = -2.612490$.

The ACT DR6 contribution is

$\Delta \chi^2_{\text{ACT}}$

$2 = -3.372038$,

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0.42

while the combined non-ACT contribution is

$2 = +0.759548.$

$\Delta \chi_{\text{non-ACT}}$

Within ACT, TE and EE contribute 1.980919 and 1.888310, while TT contributes --

$+0.497192.$

Figure 4. Signed likelihood attribution for the active $H_0 = 71.5$ EDE vector on the internally consistent serialized basis. The three ACT spectral terms contribute to the ACT total; the full-chain total combines ACT and non-ACT contributions. These bars include subtotals and must not all be summed. Values are reproduced from Chapter 4.

These numbers matter because they overturn the simplest version of an earlier story. The historical positive ACT cost at high H_0 was not a universal barrier that survived every improved basin search. At the corrected active vector, ACT can instead improve the fit relative to the serialized attribution baseline.

The lesson is broader than the sign of one contribution. The likelihood response depends on where the optimizer lands in a multidimensional parameter space. A statement such as "ACT rejects high- H_0 EDE" is therefore too coarse to survive the corrected analysis.

Varying Electron Mass

Changing the electron mass during recombination provides a physically different route.

A larger electron mass changes atomic energy scales, recombination physics, the Thomson scattering cross-section, and the distance scale inherited from the early Universe. The tested common-backend reference has

-1 Mpc

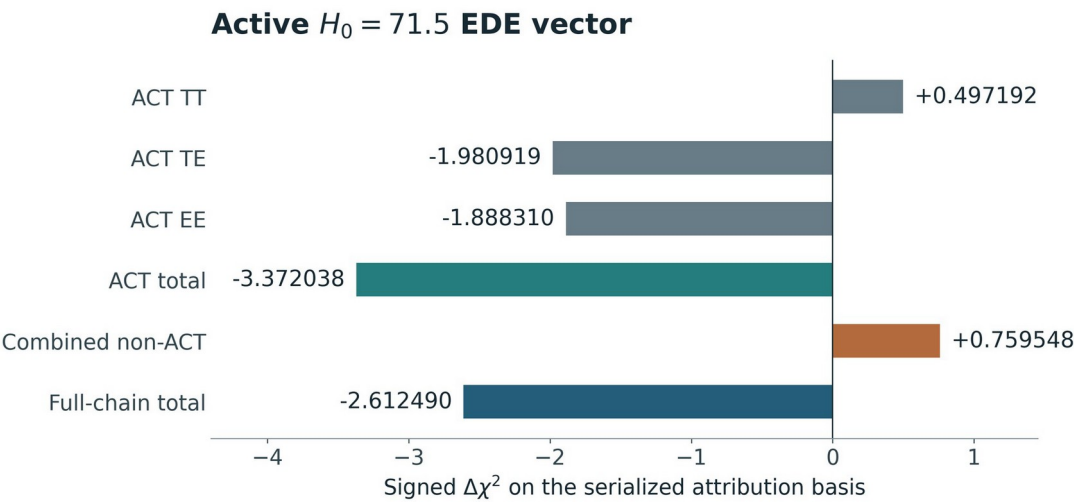
$H_0 = 70.157184 \text{ km s}^{-1}$

-1

and

$m_e/m_{e,0} = 1.0133561.$

When this mechanism is pushed to higher H_0 , the fit cost does not appear in the same place as it does for $n=3$ EDE.



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0.42

At $H_0 = 71.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$, the total tested penalty is approximately +6.81, with DESI DR2 providing the largest individual positive contribution. At $H_0 = 72.5$ and $73.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$, Big Bang nucleosynthesis becomes the largest positive contribution, while the ACT contribution becomes approximately neutral.

At the $H_0 = 73.5$ point, the tested trajectory reaches approximately

$m_e/m_e,0=1.04055$,

$\omega_b=0.02337$,

$\omega_{cdm}=0.13282$,

and

$r_d=139.38 \text{ Mpc}$.

The mechanism can therefore shrink the ruler substantially, but the compensating changes encounter pressure from other observations.

The Same H_0 Does Not Mean the Same Constraint Geometry

The comparison between $n=3$ EDE and varying electron mass reveals one of the central results of the computational programme.

Two mechanisms can move H_0 in the same direction and reduce the sound horizon to a similar scale without producing the same likelihood response.

There is no single observational component that acts as a universal high- H_0 barrier across all early-Universe mechanisms tested here.

The reason is physical. Different mechanisms obtain a smaller ruler by changing different quantities. Those changes propagate into different observables, parameter degeneracies, and likelihood components.

The scientific target is therefore not simply a preferred value of H_0 .

It is a complete parameter trajectory that survives the combined data.

CMB Likelihoods Do Not Respond Alike

The same principle appears when different CMB likelihoods are compared on a deliberately matched $n=3$ EDE surface.

Planck, SPT-3G, and ACT do not respond identically when H_0 is fixed near $71.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$.

On the matched calculation used here, Planck reaches its free- H_0 minimum at $H_0 = 67.9883 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and incurs an objective increase of +5.82386 at $H_0 = 71.5$.

SPT-3G reaches a free- H_0 value of $71.1721 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and incurs an increase of only +0.03350.

ACT reaches $71.49995 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and incurs an increase of +0.01613.

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0.42

These are separate likelihood responses. They must not be summed as though they were independent pieces of one experiment, and their numerical differences are not themselves measurements of new physics.

What they establish is narrower: the tested CMB likelihoods can have materially different high- H_0 geometry even when the cosmological model and comparison target are deliberately matched.

Why the Planck Likelihood Construction Matters

The Planck investigation was then pushed deeper.

A matched reduced likelihood and a native full-multifrequency likelihood are not the same mathematical objective. They differ in the information retained, frequency construction, covariance, foreground treatment, calibration structure, and nuisance-parameter space.

On the matched Planck surface, the fixed-minus-free objective increase is +5.8238567748.

On the tested native full-multifrequency CamSpec construction, the corresponding increase is +17.3979348723.

The difference between those two numbers is diagnostically important, but it is not an additional physical $\Delta\chi^2$ contribution. The two calculations evaluate different likelihood constructions.

Later calculations therefore examined the internal Planck likelihood architecture rather than treating the difference as one mysterious extra penalty.

Those tests found distributed structure across calibration, foreground, covariance, cosmological, and nuisance directions. No single tested component could be promoted into a demonstrated physical defect.

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0.42

Figure 6. The matched CamSpec-lite and native full-multifrequency CamSpec calculations evaluate different Planck likelihood constructions. Their fixed-minus-free objective increases therefore diagnose likelihood-architecture dependence; their numerical difference is not an additional physical $\Delta\chi^2$ contribution.

The CamSpec–HiLLiPoP Methodological Discrepancy

The analysis was later extended across two Planck PR4/NPIPE likelihood implementations, CamSpec and HiLLiPoP.

An apparent historical contradiction first emerged because an earlier CamSpec endpoint family had been labelled as stable multibasin structure, whereas later CamSpec and HiLLiPoP runs did not recover repeated stable basins under a different common-geometry classifier.

A direct classifier crosswalk resolved that binary contradiction.

When both classification rules were applied to the same preserved endpoint sets, the historical rule classified them as stable while the later graph rule classified them as non-stable.

The stable-versus-non-stable contrast was therefore materially dependent on the classifier definition. It could not by itself establish that a physical likelihood structure had disappeared.

The underlying continuous geometry difference did not disappear with that correction.

When preserved CamSpec and HiLLiPoP endpoint families were compared in the same normalized common coordinates, and later on exactly matched TT information support, a substantial implementation-dependent difference remained.

Further localization showed that this difference was distributed across several coupled coordinates rather than being reducible to one obvious parameter.

MATCHED CAMSPEC-LITE

n = 3 EDE

Matched TT + TE + EE

Matched multipole support

Fixed-minus-free objective increase

+5.8238567748

NATIVE FULL-MULTIFREQUENCY CAMSPEC

n = 3 EDE

Full frequency structure

Native covariance and likelihood

Foreground parameterization

Calibration and nuisance coupling

Fixed-minus-free objective increase

+17.3979348723

DIFFERENT LIKELIHOOD OBJECTIVES

Their difference diagnoses architecture dependence; it is not an additive physical $\Delta\chi^2$ contribution.

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0.42

The tested interventions also narrowed the possible explanation. Relative-calibration semantics alone were insufficient. Native off-diagonal precision coupling alone was insufficient. Their predefined combination was insufficient. Allowing or freezing the native foreground nuisance profiles was also insufficient to remove the discrepancy.

These are negative results, but they are informative negative results.

They rule out several simple implementation-level explanations without identifying the deeper cause.

The Common CMB-Space Numerical Boundary

The comparison was pushed one level deeper without manufacturing a synthetic hybrid likelihood.

A mathematically legitimate common physical object was defined by allowing CamSpec and HiLLiPoP to retain their own native data, covariance, foreground models, priors, and nuisance parameters while marginalizing those native nuisance parameters separately around a common physical CMB-space representation.

That construction is methodologically supported.

Its tested numerical representations were not.

A single-Gaussian Laplace/Schur approximation failed mandatory validation. A subsequent defensive randomized quasi-Monte Carlo implementation avoided that single-Gaussian assumption but also failed the preregistered numerical-stability criterion.

The endpoint stage was therefore blocked.

No validated marginalized CamSpec–HiLLiPoP endpoint geometry was produced.

This is a technical failure of the tested finite numerical representations. It is not a physical falsification of the marginal likelihood, not evidence that either likelihood is wrong, and not evidence against early dark energy.

The Downstream Portability Test

A later preregistered campaign attempted the downstream consequence directly: would the established CamSpec–HiLLiPoP fitted-geometry difference remain cosmologically material when the two Planck likelihood arms were combined with the same external information?

The executed campaign kept the $n=3$ early-dark-energy backend fixed and combined each Planck arm with ACT DR6 primary CMB information, ACT DR6 lensing, and DESI DR2 BAO in a common set of baseline and leave-one-out data combinations. Two independent chains were required for each arm and combination, producing 32 logical chains.

The computation itself reached a valid terminal state, but the inferential gates did not.

Thirty logical chains reached the frozen maximum of 20,000 accepted samples without

BUBBLEVERSE MAIN BOOK

0.42

convergence. The two remaining chains reached the eighth and final compute segment while still partial. No complete converged 32-chain posterior matrix was documented. The final state was therefore a controlled NO_SCIENTIFIC_RESULT rather than a technical crash. Chain completeness and the hard science convergence gate were blocked, so no valid standardized-shift, overlap, or leave-one-out portability classification could be issued. Non-convergence is not evidence that the two likelihoods agree, and it is not evidence that they differ physically.

A separate scope audit also matters. The executed campaign was narrower than the earlier full downstream specification: it contained the $n=3$ EDE model only, omitted separate Λ CDM control cells and the common supernova block, used three leave-one-out combinations rather than four, and applied a different decision rule. That does not invalidate the preregistered campaign that was actually run, but it means the run cannot substitute for the broader planned comparison.

The downstream cosmological significance of the fitted-geometry difference therefore remains unresolved. The computational firewall worked: where the required posterior validity was not obtained, no physical conclusion was manufactured.

Scientific Status After the Downstream Test

The result of the early-Universe programme is consequently constrained rather than triumphant.

No tested mechanism has been established as a robust complete solution to the Hubble tension.

$n=3$ early dark energy remains active but constrained. It can occupy higher- H_0 regions on the tested surfaces, but its viability depends strongly on dataset, likelihood construction, parameter basin, and implementation.

Varying electron mass also remains a constrained partial mechanism. It can reduce the early-Universe ruler and raise H_0 , but the compensating trajectory transfers fit pressure toward other constraints, particularly DESI DR2 and Big Bang nucleosynthesis in the tested profiles.

The Planck implementation studies establish a genuine methodological result: different likelihood constructions applied to overlapping Planck information can produce materially different fitted geometry.

They do not establish which likelihood is physically superior.

They do not identify a calibration error, foreground contamination, covariance defect, or instrumental failure.

They do not establish or falsify $n=3$ early dark energy.

They do not establish new physics.

The direct downstream campaign likewise did not resolve whether the fitted-geometry difference survives common external data as a material cosmological shift, collapses to

BUBBLEVERSE MAIN BOOK

0.42

effectively equivalent constraints, or depends on the dataset combination. Its controlled no-science endpoint leaves those alternatives open.

The remaining problem is narrower than it was at the beginning of the chapter, but it remains unresolved.

The technical likelihood construction, full parameter vectors, endpoint geometry, intervention tests, common-CMB numerical diagnostics, downstream convergence record, and complete provenance are preserved in Appendices A–C.

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0.42

Chapter 5 — What the Model Explains— and What Remains Unknown

Dark Energy Is an Explanation, Not a Detected Substance

The same epistemic care applied to dark matter and the Hubble constant also applies to dark energy.

Supernovae and large-scale-structure distance measurements strongly support late-time accelerated expansion within the relevant relativistic cosmological framework. Λ CDM represents the cause of that acceleration through a cosmological constant.

Other models allow the effective dark-energy contribution to evolve.

DESI DR2 combinations with CMB and Type Ia supernova data provide model-dependent evidence for time-varying dark-energy parameterizations over a pure cosmological constant. The original DESI DR2 analysis reported preferences ranging from approximately 2.8σ to 4.2σ depending on the supernova compilation. Subsequent 2026 reanalyses using updated host-galaxy information and recalibrated supernova samples bring the comparisons into substantially closer agreement, at approximately 3.2σ – 3.4σ in the w_0w_a plane.

This is scientifically interesting model-comparison evidence.

It is not a direct observation of a changing dark-energy field.

The result depends on the datasets being combined and on the parameterization used to describe the possible evolution.

The physical origin of late-time acceleration therefore remains unresolved. A cosmological constant remains viable, and dynamical dark energy remains an active possibility.

What Λ CDM Actually Achieves

Λ CDM deserves neither automatic belief nor casual dismissal.

Its empirical success is substantial.

With a small number of parameters, it provides a compact description of the cosmic microwave background temperature and polarization spectra, the acoustic peak structure, baryon acoustic oscillations, important aspects of cosmic geometry, matter density, and the broad development of large-scale structure.

Independent CMB experiments recover compatible features over overlapping observational ranges, strengthening the conclusion that the measured structures are not artefacts of one instrument.

The correct scientific statement is therefore not that Λ CDM has been directly observed.

It is that Λ CDM is an extraordinarily successful model of a large body of observations.

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0.42

A successful model is evidence about nature.

It is not identical to nature.

What the Evidence Establishes

The observational foundation developed in Chapter 1 remains intact.

The observable Universe is structured and evolving. Cosmological redshifts, distance relations, baryon acoustic structure, the cosmic microwave background, primordial-abundance information, gravitational lensing, gravitational waves, and other independent observational channels jointly support a Universe that was once much hotter and denser and whose large-scale geometry evolves with time.

Late-time acceleration is strongly supported within the relevant cosmological framework.

The combined gravitational evidence requires either a substantial additional non-luminous gravitating component within the standard gravitational description or an alternative gravitational theory capable of reproducing the same range of phenomena.

The Hubble tension is also real in the specific sense established earlier in this book: materially different inference chains for the same parameter remain in conflict. Chapters 2 and 3 give the full numerical benchmarks, uncertainties, dependencies, and comparisons. The local distance network, the combined CMB inference, and the DESI DR2 plus Big Bang nucleosynthesis standard-ruler route remain distinct inference chains rather than three identical instrument readings..

What Remains Unknown

Several of the most important physical questions remain open.

The microscopic identity of dark matter is unknown.

The physical origin of late-time acceleration is unknown.

The Hubble tension remains unresolved.

No tested early-Universe mechanism in this investigation has been established as a complete solution to that tension.

The full size and topology of the Universe are not determined by observations within our causal region.

The physics of epochs earlier than those directly constrained by the cosmic microwave background and nucleosynthesis becomes progressively more model dependent.

The Planck likelihood investigation has also left a methodological problem unresolved.

CamSpec and HiLLiPoP produce materially different fitted parameter geometry on overlapping Planck information under the tested constructions. Several simple implementation-level explanations have been rejected as sufficient, but the causal implementation origin has not been isolated.

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0.42

A mathematically legitimate route was established for comparing the two likelihoods after marginalizing their native nuisance parameters separately, but the tested finite implementations did not pass the required numerical validation and therefore produced no final marginalized endpoint geometry.

A later direct downstream MCMC campaign also terminated without a scientific classification because the required posterior convergence was not achieved within its frozen sample and segment budgets.

That later campaign was narrower than the broader planned Λ CDM-versus-EDE comparison with a common supernova block, so the complete downstream scientific question remains open. Both limitations are computational or methodological; neither is a physical result.

The Current Scientific Position

The current Bubbleverse state is therefore deliberately incomplete.

The observational foundation is strong.

Λ CDM remains a highly successful compact model.

The Hubble tension remains a robust parameter-inference conflict whose cause is unknown.

$n=3$ early dark energy remains active but constrained rather than established.

Varying electron mass remains a constrained partial mechanism rather than an established reconciliation.

The CamSpec–HiLLiPoP comparison remains a methodological likelihood-implementation problem, not an independent disagreement between observations of the sky.

The attempted downstream portability campaign did not determine whether this methodological geometry difference becomes a material cosmological inference shift, collapses under common external data, or is conditional on the particular external dataset combination.

No result in the surviving record establishes a specific dark-matter particle, a microscopic dark-energy mechanism, a physical Big Bang singularity, a preferred Planck likelihood as physical truth, or new physics as the cause of the Hubble tension.

This boundary between strong evidence and unfinished explanation is not a defect in the model.

It is the point of the model.

The final chapter now turns from what Bubbleverse currently has the right to say to how its principal numerical conclusions were actually produced, corrected, validated, and preserved.

BUBBLEVERSE MAIN BOOK

0.42

Chapter 6 — Experimental and Computational Record

Purpose of the Computational Programme

The Bubbleverse computational programme answers a narrower question than the preceding scientific chapters: what was actually run, what did each calculation test, and how securely can its result be reconstructed?

A successful GitHub Actions execution shows that a specified computational procedure ran. It is not, by itself, new observational evidence. A failed workflow is likewise a computational failure unless a valid scientific calculation had already completed before the downstream failure. Execution status and scientific status must therefore remain separate. The programme developed progressively. Early runs established a controlled common backend for comparing candidate high- H_0 mechanisms. Later campaigns tested optimizer sensitivity, corrected superseded minima, decomposed likelihood responses, compared Planck, SPT-3G, and ACT on deliberately matched surfaces, examined full-multifrequency Planck likelihood structure, and tested the portability of preserved CamSpec endpoint geometry under HiLLiPoP and later CamSpec reconstructions.

That process also exposed an important methodological lesson. Binary basin labels depended on the classifier used, while a continuous CamSpec–HiLLiPoP geometry difference remained after the classifier semantics were harmonized. The surviving discrepancy is therefore methodological likelihood evidence, not an independent disagreement between observations of the sky and not a detection of new physics...

Repository, Software, and Frozen Scientific Configuration

The primary Bubbleverse computational repository is Morfindien/Bubbleverse. The reconstruction performed for this edition uses the repository's main branch as the principal execution record.

The frozen $n=3$ early-dark-energy calculations repeatedly depend on mwt5345/class_ede at commit 5a131c91d657dd9a7c6364cc45b038710f8d0d97.

The ACT DR6-lite calculations repeatedly depend on ACTCollaboration/DR6-ACT-lite at commit 0e0cd2c703c62a0e980470b572602233b27750e1.

These commits are preserved as computational provenance rather than treated as generic software labels.

The documented common-backend environment included Python 3.10.x, Cobaya 3.5.6, NumPy 1.26.4, SciPy 1.15.3, Py-BOBYQA 1.5.0, Cython 0.29.37, and SACC 1.0.2. Individual later workflows record more specific runner and Python versions where relevant.

The central methodological constraint was preservation of the scientific surface. Later robustness, attribution, matching, and precision calculations were not permitted to become

BUBBLEVERSE MAIN BOOK

0.42

unrecorded changes of model, dataset, prior, or likelihood. Where a later calculation changed only numerical search strategy, validation semantics, arithmetic representation, or the way an existing result was aggregated, that distinction was retained explicitly.

Common-Backend Model Comparison and Globality Recovery

The early common-backend programme established a controlled environment in which $n=3$ early dark energy and varying electron mass could be compared without pretending to reproduce the native pipelines of the external papers exactly.

Repeated common-backend runs are recorded under GitHub Actions run Ids 33249784822 and 33252632548 at repository commit b875ea95d9b6390b8f03dac729eb7f5fe84d3118.

A subsequent deterministic globality-recovery calculation was executed as GitHub Actions run 33271326245 through the Bubbleverse Q005 V14 Globality Recovery workflow at execution commit 7293c7f5db4d7ab72b1aaad38fe15f2639e11d5e.

The corresponding result is R-Q005-V14-GLOBALITY-RECOVERY-001. The result JSON SHA-256 is 3642aa6e4d09c595610875cfed321f3d28055fb059757c2381f08007bf071253.

The purpose of this stage was not to claim a paper-native reproduction. It was to establish a controlled common-backend comparison and determine whether the numerical conclusions were sensitive to optimizer-basin discovery.

That sensitivity became scientifically important.

Globality, Corrected Minima, and ACT Attribution

The fixed- H_0 early-dark-energy profiles were subsequently subjected to broader multi-start testing. The investigation retained H_0 targets at 70.853292, 71.5, 72.5, and 73.5 $\text{km s}^{-1} \text{Mpc}^{-1}$ while expanding the numerical search for lower basins.

The authoritative expanded globality calculation is R-Q011-EDE-GLOBALITY-

CERTIFICATION-004.

It was executed through the Bubbleverse Q011 EDE Globality Certification V4 workflow as GitHub Actions run 33363067156 at repository commit e113fb846948f424b865d1303fba8b5988ec9133.

The run used the frozen Q005 V14 scientific configuration together with the preserved class_edc and ACT DR6-lite commits.

The resulting classification was BEST-OBSERVED MINIMUM ONLY.

This classification is important. The multi-start programme improved the observed likelihood envelope, most materially at $H_0=71.5 \text{ km s}^{-1} \text{Mpc}^{-1}$, but it did not establish a mathematical proof that no lower basin exists. The numerical state therefore became better characterized without becoming mathematically global.

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0.42

The corrected $H_0=71.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ best-observed solution has optimizer-native $\chi^2=546.364260$. Relative to the optimizer-native reference $\chi^2=548.010401$ at $H_0=70.853292 \text{ km s}^{-1} \text{ Mpc}^{-1}$, this corresponds to $\Delta\chi^2=-1.646141$.

A separate serialized fixed-vector attribution uses a different internally consistent numerical reference. It gives $\Delta\chi^2=-2.612490$ at $H_0=71.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$, with ACT DR6 contributing -3.372038 and the combined non-ACT terms contributing +0.759548. The difference between optimizer-native and serialized bookkeeping is preserved as a finite-precision round-trip issue rather than redistributed as a physical likelihood contribution. The corrected ACT fixed-point attribution is recorded as INTERNAL-Q009, logical run Q009-

ACT-CORRECTED-MINIMA-ATTRIBUTION-V8.

The calculation used `q009_act_corrected_minima_attribution_v8.py` with `q009_act_corrected_minima_attribution_v8_config.yml`. The repository inspection state was commit `a8d11cc9687e5df757f4794224d83ed34df03846`.

The calculation evaluated complete selected best-fit vectors without introducing a new optimizer, sampler, prior, parameter bound, or dataset.

The exact successful GitHub Actions run identifier for this Q009 execution is not established by the surviving Word provenance and is therefore not invented here.

Planck-SPT-ACT Comparative Programme

The next computational stage tested whether the behaviour found on one CMB likelihood generalized across experiments.

The Q015 covariance-aware Planck/SPT attribution is recorded as R-Q015-EDE-CMB-

ATTRIBUTION-003.

The scientific endpoint calculations were produced by the Q015 V2 workflow at repository head `78a65aae897fca18132a5c0758bccdbbc6a09402` in GitHub Actions run `33629190153`.

The overall V2 workflow concluded with failure, but that status must not be confused with failure of every scientific calculation inside it. The Stage-0 frozen-parent job completed successfully. The Planck endpoint job completed successfully. The SPT endpoint job completed successfully. Their artifacts were produced before the downstream merge and mandatory-test job failed.

The preserved Stage-0 artifact is `q015-v2-stage0`, artifact ID `9846055938`, SHA-256 `7adb0624a73f8071f2b4dfcac6e65ebfe2d4076277156a87b090c0cab7f99696`.

The preserved Planck endpoint artifact is `q015-v2-endpoint-planck_npipe_k039_approx`, artifact ID `9846143033`, SHA-256

`ff6439352ae00ad0c87befb07b035342abe5d1fecf5dee09b4eb84f241bad3a6`.

The preserved SPT endpoint artifact is `q015-v2-endpoint-spt_d1_only`, artifact ID

9846127249, SHA-256

`b0740d6424b71fb85a6adf92a9bbd2205b27479fec2e7d2f6d97ee8c2be0c33`.

BUBBLEVERSE MAIN BOOK

0.42

The later V3 certification changed the validation semantics required to merge the already completed endpoint calculations. It did not change the scientific model, endpoints, likelihoods, covariance matrices, theory calculation, optimizer results, or signed attribution values. The final Q015 V3 GitHub Actions run identifier remains a provenance item requiring verification before publication.

The matched Planck–SPT–ACT analysis then placed the three experiments on a deliberately restricted common comparison: the same $n=3$ model, TT, TE, and EE observables, and the multipole range $600 \leq \ell \leq 2000$, while retaining each likelihood's native nuisance structure. $\leq$
The matched residual-mechanism analysis is R-Q016-EDE-CMB-MECHANISM-004, logical run Q016-CMB-MECHANISM-V4.

The successful GitHub Actions execution was run 33836015653 at repository head 91bc0736a00e2454d4e37324e5d3a1a7fe7813d6.

The matched comparison found a much larger high- H_0 response from Planck than from SPT-3G or ACT on the tested surface. The individual likelihood objectives remain separate. No combined Planck–SPT–ACT significance was constructed, and shared cosmic variance was not represented through a unified cross-experiment covariance.

The calculation therefore localized a difference in likelihood response. It did not establish that any experiment was wrong.

Planck Likelihood Architecture and Preferred-Region Geometry

The Planck-specific programme then asked whether its high- H_0 response could be reduced to a simple calibration, foreground, frequency, or parameter direction.

The full-multifrequency nuisance-direction analysis is recorded as R-Q017-EDE-PLANCK-DIRECTION-LOCALIZATION-003, logical run Q017-PLANCK-DIRECTION-LOCALIZATION-V3.

The successful GitHub Actions execution was run 33842355384 at repository commit 277bb274e5783831303b542681e8cf87a70d31c2.

The final artifact is q017-final-v3, artifact ID 9925585139, SHA-256 a45ba197f20595013d390136e00f8fd83f40a70fb9e9e1d83f2048f2d70320db.

The authoritative classification was INDETERMINATE between a specifically localized and a broadly distributed Planck penalty. Calibration was an important tested nuisance direction, but it was not identified as the physical cause.

The likelihood-bridge programme then demonstrated that the matched CamSpec-lite and native full-multifrequency CamSpec constructions could yield substantially different high- H_0 penalties even when evaluating the same $n=3$ model. These objectives are not additive physical χ^2 contributions.

Independent reprofiling further showed that similar projected values of H_0 and f_{EDE} do not imply equivalent multidimensional preferred regions.

The authoritative preferred-region result is R-Q019-EDE-PLANCK-COSMOLOGY-

REPROFILE-003.

BUBBLEVERSE MAIN BOOK

0.42

The earlier R-Q019-EDE-PLANCK-COSMOLOGY-REPROFILE-001 interpretation is superseded, while its valid endpoint and cross-profile calculations remain preserved as historical numerical provenance.

The subsequent block-localization analysis is R-Q020-EDE-PLANCK-CROSS-PROFILE-DIRECTION-002, logical run Q020-PLANCK-CROSS-PROFILE-DIRECTION-REPAIR-V2.

The successful GitHub Actions execution was run 33878880397 at repository commit b0c0a8fd89408bac91502e15d42c69820e913ab6.

The final artifact SHA-256 is

9fed5d755abdb3630bbe3a1a88e23b0c8172b49dab129a4421b527c3bdabde7f.

Within its predefined four-block representation, the full-multifrequency response was dominated functionally by the primordial block, but the resulting Shapley allocations and interactions are diagnostics of a correlated likelihood surface rather than independent physical components.

The primordial block was subsequently opened into n_s , $\log A$, and τ_{reio} .

The authoritative result is R-Q021-EDE-PRIMORDIAL-INTERNAL-STRUCTURE-002.

The original Q021 numerical execution produced 48 finite primordial-profile evaluations across two likelihood constructions, eight primordial vertices, and three optimizer restarts. Its original final aggregation was later found to mix independently selected best vertices from different optimizer restarts, creating a decomposition that did not correspond to one coherent optimization trajectory.

The repair did not discard the valid raw likelihood evaluations.

Instead, it rebuilt coherent restart-specific decompositions, replaced a process-dependent likelihood identity representation with a stable structural identity, and prohibited cross-restart mixing. No new likelihood evaluations were required for that repair.

All coherent decompositions retained the qualitative classification that the primordial response is distributed rather than reducible to n_s alone.

Full-Multifrequency Basin Programme

The next campaign followed the full-multifrequency endpoint structure itself. The historical continuation result is R-Q022-EDE-FULLMF-OPTIMIZER-BASIN-002 at execution commit 16a7902c3be7d1967adaae9ee380baf0d17e903a. That execution recentered finite scalar Cobaya references on the selected endpoint values for the sampled non-primordial parameters while holding n_s , $\log A$, and τ_{reio} fixed at each primordial vertex. The resulting endpoint families remained separated and were classified at the time as stable numerical multibasin structure. The later full-native CamSpec revalidation, which also used scalar references but sampled the primordial block, did not reproduce that classification. A subsequent provenance audit established that reference representation itself was therefore not the differing variable. The historical endpoint vectors and objective values

BUBBLEVERSE MAIN BOOK

0.42

remain valid outputs of their original parameterization, while the stronger interpretation of the basin classification as parameterization-independent is not supported.

The subsequent fixed-lineage direction analysis is R-Q023-EDE-FULLMF-BASIN-

DIRECTIONS-002.

It was executed in GitHub Actions run 33913077353 at execution commit ea5af9a0b25090eb8472adb3c9574dcb4e1efe5c.

The permutation- and sign-invariant correspondence test is R-Q024-EDE-FULLMF-BASIN-

MATCHING-001.

It was executed in GitHub Actions run 33915600167 at execution commit b8e16629bbc6592ee1f0d90f78c887a913ac4c1.

It tested all 36 joint relabellings without introducing new likelihood evaluations and did not recover a universal basin-separation direction under the predefined criterion.

The fixed-vector component attribution is R-Q025-EDE-FULLMF-BASIN-COMPONENT-ATTRIBUTION-002, logical run Q025-FULLMF-BASIN-COMPONENT-ATTRIBUTION-V2.

The successful GitHub Actions execution was run 33918779905 at execution commit b2db6827b1caf5405e6a34bff221b6ae4813c1d0.

The final artifact is q025-final-v2, artifact ID 9954333941, SHA-256 104b584c9212a65e7f06cae64a732dac6f58b801cb8feec34c7f49d1bfddac0f.

This successful V2 execution completed all nine required basin edges and all 72 fixed-vector likelihood evaluations, with zero optimizer evaluations.

The historical Q025 V1 execution must remain separate. GitHub Actions run 33917809217 failed because the Q022 endpoint serialization was read incorrectly. It completed zero of the required 72 likelihood evaluations and therefore carries no scientific evidentiary weight.

The subsequent coordinate-level localization is R-Q026-EDE-FULLMF-MASK6-COUPPING-001, logical run Q026-MASK6-NUISANCE-COUPPING-LOCALIZATION-V1.

The successful GitHub Actions execution was run 33921255497 at execution commit 7cb4d98c01196c22b407b6f4fd4118179afea3de.

It completed 252 of 252 required fixed-vector likelihood evaluations and performed no optimization. Within the second tested mask, the distinctive compensation localized strongly around A_{planck} and its coupling to cosmological coordinates, particularly $\omega_{\text{cdm}} \times A_{\text{planck}}$. This remained functional likelihood geometry rather than evidence of a calibration failure.

The residual-and-covariance bridge is R-Q027-EDE-FULLMF-RESIDUAL-COVARIANCE-

BRIDGE-001.

The successful GitHub Actions execution was run 33924573651 at execution commit df03e30712690e61ff799558598405a22c0bacbb.

BUBBLEVERSE MAIN BOOK

0.42

The selected TT frequency region at $600 \ell < 1000$ was shown to be insufficient as a unique $\leq$ localization of the interaction.

The common-support counterfactual is R-Q028-EDE-FULLMF-COMMON-SUPPORT-COUNTERFACTUAL-008, logical run Q028-FULLMF-COMMON-SUPPORT-COUNTERFACTUAL-

V8.

The successful GitHub Actions execution was run 33957937309 at execution commit 7f5d35c0725530039162adfa4d71be288e8b4462.

The validated exact labelled common support contains all 9915 dimensions. The calculation completed its fixed-vector and checkpointed work without optimizer or sampling evaluations and produced the final classification

VALID_COMMON_SUPPORT_SEPARATION.

The final artifact is q028-final-v8, artifact ID 9967068363, SHA-256

7dfbd1f27251ff311dd1de6118539dd691ab03d9afdb08539b3b13586c50d5af.

An earlier execution SHA and common-support hash were subsequently identified as provenance misrecords and are superseded. They must not be restored as active identifiers.

Residual Geometry, Common Support, and Exact Interaction Audit

The common-support result allowed the relevant interaction to be expressed exactly as $2 (m^{\text{TPm}})$.

$2m + \Delta$

$F = -2d^{\text{TP}} \Delta$

The pure quadratic data term cancels because the same data vector appears in all four factorial states. The first surviving term is a data-model factorial coupling; the second is a model-quadratic response. The decomposition therefore does not provide a unique physical split into data-driven and model-driven causes.

The full nine-case numerical diagnostics are preserved in Appendix B.

The final numerical precision audit is R-Q030-ORIGINAL-GATE-PRECISION-AUDIT-001.

The validation history matters because several kinds of failure occurred before the final certification.

An initial Q030 execution failed before a scientific result because it used a misrecorded common-support hash.

GitHub Actions run 33961469874 then produced all nine diagnostics but failed the original numerical closure gates at approximately 10^{-8} to 10^{-7} because cancellation-sensitive differences of large quadratic scalars were being evaluated directly.

A later validation-only run, 33961879699, accepted the unchanged values under a relaxed 10^{-7} numerical threshold. That run is not the authoritative final validation.

The one-shot precision audit in GitHub Actions run 33962896373 demonstrated that the original 10^{-8} gate could instead be satisfied using algebraically equivalent, numerically

BUBBLEVERSE MAIN BOOK

0.42

stable arithmetic. Its workflow nevertheless failed because the validator looked for an incorrect schema key.

The finalization repair was then executed through the Bubbleverse Q030 audit finalization validator repair R1 workflow.

The successful GitHub Actions execution was run 33963221940 at head commit 4cbe1585d2603b507679c654a8ff8b0d7e2e522e.

The final artifact is q030-original-gate-precision-audit-final-v1, artifact ID 9968592047, SHA-256 7685d7693f2e85391767bd9a2654a940280a02c729ff5a441b988c69a752a156.

The finalization did not rerun the scientific audit. It did not change a mask shard, a scientific input, a numerical value, or a tolerance. It repaired the validator and reconstructed the final result from the immutable audit artifacts.

The stable factorial-bilinear representation reaches a maximum identity closure error of $2.0940194023211234 \times 10^{-13}$, a maximum spectrum- or multipole-pair aggregation error of $5.479297571220343 \times 10^{-14}$, a maximum $|\Delta^2 m + \Delta^2 r|$ of $4.547473508864641 \times 10^{-13}$, and a maximum parent-functional closure error of $1.6143639203214377 \times 10^{-10}$.

These satisfy the original validation thresholds.

Failed, Repaired, and Superseded Executions

The computational history cannot be reduced to a sequence of green check marks.

Some failures occurred before any scientific evaluation and therefore produced no evidentiary result. The Q025 V1 endpoint-serialization failure is one such case.

Some workflows produced scientifically valid child calculations but failed during aggregation or validation. Q015 V2 belongs to this class: its Stage-0, Planck, and SPT scientific jobs succeeded and produced preserved artifacts, while the downstream merge job failed.

Some later executions did not rerun the science at all. They repaired validation semantics, aggregation, identity checks, provenance, or arithmetic representation while preserving the underlying scientific evaluations. The repaired Q021 aggregation and the final Q030 validator repair belong to this class.

Other calculations were scientifically superseded because broader optimization found better parameter basins. The historical $n=3$ EDE fixed- H_0 attribution remains valid for the parameter vectors at which it was evaluated, but it is not the active description of the corrected best-observed basins.

Supersession therefore does not mean deletion of history.

It means that the historical result must be labelled with the state to which it actually applies.

BUBBLEVERSE MAIN BOOK

0.42

Downstream Portability Campaign

The next major execution attempted to propagate the validated CamSpec–HiLLiPoP fitted-geometry difference into matched downstream n=3 EDE inference. The authoritative implementation is Q041-PLANCKPORT-V19, result R-Q041-EDE-DOWNSTREAM-

PORTABILITY-019.

The V19 matrix contained two Planck arms, eight EDE data combinations, and two independent chains per arm/combination. The external blocks were ACT DR6 primary CMB (A6), ACT DR6 lensing (L6), and DESI DR2 BAO (D2). The Planck parent was the validated

Q032 exact-common-TT construction; with ACT primary present, the overlap policy

restricted Planck primary TT support to $\ell \leq 599$ while ACT primary TT/TE/EE began at $\ell = 600$.

Cobaya MCMC used `Rminus1_stop = 0.02`, `Rminus1_cl_stop = 0.2`, a hard science R-hat gate of 1.05, a maximum of 20,000 accepted samples per chain, and at most eight successful compute segments per chain. Each segment had a 300-minute soft stop inside a 330-minute GitHub job timeout. Same-run artifacts transported the immediate predecessor chain state; cross-version state reuse was forbidden.

GitHub Actions run 34979609004 completed successfully on attempt 4 at execution commit a962ba70077422f8b69642dea4e8272e1d00e5ff. The final artifact q041-final-v19, artifact ID 10754769330, has SHA-256

2284518ecbaaf1ff0d0093977827d5162ec29549ba2b81880f266c250311ca2b.

The scientific matrix did not converge. Thirty chains were reported as MAX_SAMPLES_WITHOUT_CONVERGENCE. The two remaining chains were still PARTIAL after the eighth compute segment. The final object therefore records `scientific_classification = NO_SCIENTIFIC_RESULT`, `outcome_type = CONTROLLED_NO_SCIENTIFIC_RESULT`, `technical_failure = false`, `final_outcome_valid = true`, while CHAIN_COMPLETENESS, ALL_RHAT_LE_1P05, and BOTH_ARMS_ALL_COMBINATIONS are blocked.

The green workflow conclusion must therefore be read as successful enforcement of the no-science firewall, not as a successful cosmological inference. No material, equivalent, or mixed downstream class was obtained.

A provenance audit also found that V19 is not identical to the broader downstream design that preceded it. V19 contains no separate Λ CDM control cells or common supernova likelihood, uses three leave-one-out combinations, and applies a different classification rule. The difference is a scope distinction, not a retroactive invalidation of V19, but it prevents V19 from closing the broader planned test.

Reproducibility Status

The computational record has several reproducibility classes. The Q005 globality-recovery calculation, Q011 globality certification, Q016 matched CMB mechanism analysis, Q017 full-multifrequency localization, Q020 cross-profile direction analysis, the historical Q022

BUBBLEVERSE MAIN BOOK

0.42

continuation, the Q023–Q028 fixed-endpoint programme, the Q030 final audit and finalization, the Q031 HiLLiPoP portability test, and the authoritative Q032 full-native CamSpec revalidation have sufficiently explicit repository, workflow, run, commit, result, and validation lineage in the surviving record to support strong reconstruction of their execution histories. The Q033 provenance audit adds a source-locked reconstruction of the

Q019 Q021 Q022 and Q032 parameter-reference semantics. The authoritative audit is — —

R-Q033-EDE-LAUNCH-SEMANTICS-AUDIT-002, GitHub Actions run 34055678308, head SHA 614b227c8de5ea2096f75bf8b97155f340cb1a44. The audit corrects the interpretation rather than the raw computations: the historical Q022 and Q032 paths both used scalar pointlike references, while their explicit sampled-parameter sets differed by n_s , $\log A$, and τ_{reio} . The audit corrects the interpretation rather than the raw computations: the historical Q022 and Q032 paths both used scalar pointlike references, while their explicit sampled-parameter sets differed by n_s , $\log A$, and τ_{reio} . The Q015 V2 scientific endpoints are also strongly reconstructable because the workflow, head commit, successful child jobs, and endpoint artifacts are preserved even though the overall workflow concluded with a failed merge. Several older links remain only partially reconstructable from the present publication record; those are provenance gaps, not permission to invent identifiers. Other links remain only partially reconstructable from the present publication record. The successful GitHub run and head commit for the historical Q007 corrected execution are not yet established in the manuscript. The final authoritative Q008 workflow run, execution commit, and result-file hash require verification. The exact successful GitHub Actions run for the Q009 corrected-minima attribution has not been established here. The final Q015 V3 GitHub Actions run identifier also remains to be verified.

Those gaps are provenance gaps.

They are not permission to invent identifiers.

The downstream V19 campaign is also strongly reconstructable: its preregistration, source lock, scientific program, validator, workflow, execution commit, run ID, final artifact ID, and artifact digest are all preserved. Its reconstructable result is a controlled no-science endpoint; there is no converged posterior classification to reproduce.

What the Computational Programme Established

The computational programme established that the numerical structure behind the tested high- H_0 early-universe solutions is substantially more complicated than a one-parameter shift in H_0 or a single experiment acting as a universal barrier. The $n=3$ early-dark-energy likelihood surface is sensitive to optimizer discovery, and corrected best-observed high- H_0 vectors materially changed earlier ACT attribution patterns. Planck, SPT-3G, and ACT respond differently even when compared on deliberately matched observables and multipoles. The Planck response itself also depends on likelihood construction: similar projected values of H_0 and f_{EDE} do not guarantee equivalent multidimensional preferred regions.

BUBBLEVERSE MAIN BOOK

0.42

Within the historical full-multifrequency CamSpec execution, separated endpoint families involved both cosmological and nuisance coordinates. Fixed-vector analysis of those preserved endpoints found mask-dependent direction structure, an important local A_{planck} -centred compensation involving ω_{cdm} in one mask, and strongly covariance-coupled residual and prediction geometry. Stable numerical arithmetic reproduced the exact interaction diagnostics while satisfying the original validation gates. These results remain valid descriptions of the historical endpoint family.

What changed is the status of the stable-basin interpretation. The historical endpoint continuation produced separated families while the primordial block n_s , $\log A$, and τ_{reio} was fixed, and classified them using a per-mask mixed cosmology–nuisance non-collapse rule.

The independent HiLLiPoP test and the later full-native CamSpec programme instead used a common-geometry graph classifier and recovered no reproducibly supported stable basin. Restoring the omitted TT support, TE, or EE separately did not restore a stable cluster; neither did any tested pairwise combination or the full-native SUPPORT + TE + EE control.

Source reconstruction then established that both CamSpec execution paths already used scalar pointlike parameter references. A direct matched primordial-freeze test subsequently fixed n_s , $\log A$, and τ_{reio} inside the later full-native construction and still produced no stable repeated cluster under the later classifier. The nine refined frozen endpoints instead remained nine singleton clusters.

Primordial freezing is therefore insufficient to restore the historical stable-basin label within the later classification framework. The remaining apparent contradiction was subsequently resolved at the level of classification itself: when both the historical mixed cosmology–nuisance rule and the later seven-coordinate graph rule are applied to the same preserved endpoint families, the labels move together. The historical-versus-later stable-basin contrast is therefore classifier dependent rather than evidence that an invariant basin structure disappeared.

That correction does not erase the underlying cross-implementation difference. A direct continuous comparison in the same seven normalized coordinates finds materially different CamSpec and HiLLiPoP endpoint locations and internal shapes. More importantly, the difference remains when the comparison is restricted to the exact TT information support shared by the two implementations. In that matched-support analysis, all nine paired endpoint displacements exceed the locked 0.10 threshold, as do all 36 pairwise geometry drifts.

A subsequent deterministic localization analysis then examined every non-empty subset of the same seven coordinates. The result is not consistent with a one- or two-coordinate concentration. No subset containing three or fewer coordinates satisfies the complete preregistered capture and robustness requirements. The unique smallest qualifying subset contains ω_b , ω_{cdm} , $f E D E$, and θ_{scf} , and it reproduces more than 87% of the full discrepancy

BUBBLEVERSE MAIN BOOK

0.42

in each of the matched-location, centroid, and pairwise-shape channels while passing all leave-one-label-out tests.

The computational programme therefore supports a narrower but more informative methodological conclusion than the original binary portability claim. The tested CamSpec and HiLLiPoP constructions produce materially different fitted geometries; the difference survives exact TT support matching and is expressed through a distributed multivariate structure rather than a single dominant common coordinate. The tested explanations based on classifier semantics, scalar-versus-distribution launch references, primordial freezing, unequal TT information support, and simple one- or two-coordinate concentration are therefore insufficient to account for the complete discrepancy.

Its causal implementation origin remains unresolved. No physical Planck systematic, calibration error, foreground failure, covariance defect, instrumental problem, independent observational discrepancy, or falsification of $n=3$ early dark energy has been established.

Later preregistered intervention tests narrow that unresolved implementation problem further. Relative-calibration semantics were neutralized without removing the common-support geometry difference. The native off-diagonal precision coupling was then removed, again without collapsing the discrepancy. Applying both interventions together produced no compensating cancellation and instead worsened all three aggregate geometry measures. In every case, all nine leave-one-label-out tests remained materially discrepant. A separate eighteen-profile calculation then removed the freedom of each likelihood's native TT foreground nuisance parameters to vary from label to label. Each implementation's foreground vector was fixed to its own within-implementation reference endpoint, without mapping nuisance parameters between CamSpec and HiLLiPoP. The resulting median matched displacement was 2.9829, the centroid displacement was 1.0555, and the median pairwise-shape drift was 1.6353, compared with the locked common-support baseline values of 2.9318, 0.8311, and 1.6368. None of the nine leave-one-label-out analyses satisfied the predefined 0.10 criterion.

The computational record therefore excludes several simple interpretations of the implementation dependence under the operations actually tested. Relative calibration alone, native off-diagonal precision coupling alone, their predefined combination, and native foreground nuisance profiling freedom alone are all insufficient to remove the common-support geometry difference. The remaining distinction lies deeper in the coupled construction of the two likelihoods, including foreground-model form, detector-set and cross-spectrum construction, weighting and coaddition, data-vector definition, covariance propagation, masks, calibration structure, and other implementation choices that cannot always be separated into independent switches.

Later work showed that this methodological boundary does not force the analysis to manufacture a synthetic likelihood. A common physical object can instead be defined by marginalizing each implementation separately over its own native nuisance parameters while retaining the CMB TT spectrum and $A_{\text{Pl}} n_{\text{s}} k$ as the common physical coordinates. This

BUBBLEVERSE MAIN BOOK

0.42

preserves the native CamSpec and HiLLiPoP data, covariance, foreground models, and priors rather than exchanging them between pipelines.

Two numerical representations of this common-latent construction were tested. A single-Gaussian Laplace/Schur compression failed validation because the CamSpec nuisance solution encountered a native hard boundary with a non-positive-definite nuisance Hessian, while the HiLLiPoP CMB representation failed the predefined multistart-stability criterion. A subsequent non-Gaussian defensive randomized quasi-Monte Carlo calculation avoided both the single-Gaussian assumption and arbitrary mode selection. Its complete finite campaign reached the preregistered hard cap of $m=16$, corresponding to 393,216 nuisance nodes per replicate and implementation, but still failed the required numerical-stability tests. At the hard cap, the maximum transition changes were approximately 16.59 for CamSpec and 61.13 for HiLLiPoP, while the independent-bank differences were approximately 28.44 and 63.11, respectively, against a required tolerance of 0.05. The scientific endpoint stage was therefore blocked exactly as preregistered. No marginalized-likelihood CamSpec–HiLLiPoP endpoint geometry was produced, and the calculation cannot be used to claim either that the deeper structural bridge succeeds or that it fails physically. The authoritative classification is a numerical-validation failure of the tested finite representation, not a falsification of the underlying marginal likelihood and not a physical result.

The strongest surviving conclusion is consequently precise. CamSpec and HiLLiPoP produce a reproducible, distributed implementation-dependent fitted geometry on matched Planck TT support. Several simple implementation-level explanations have been rejected as sufficient, and a mathematically legitimate route to deeper nuisance marginalization has been identified, but the tested numerical implementations were not adequate to determine whether that deeper construction would materially reduce the discrepancy. Its causal origin therefore remains unresolved.

These results remain methodological computational evidence derived from overlapping Planck PR4/NPIPE information. They do not constitute independent observations, establish a calibration defect, identify foreground contamination, demonstrate a covariance or instrumental failure, determine that either likelihood is superior, favour or falsify $n=3$ early dark energy, or provide evidence for new physics.

The later downstream campaign adds one further validated computational result: under the frozen V19 sampler and compute budget, the required complete converged posterior matrix was not obtained, and the preregistered science firewall correctly blocked interpretation. This constrains the numerical route; it does not select a physical explanation.

[Internal computational provenance: Q041-PLANCKPORT-V19; R-Q041-EDE-DOWNSTREAM-PORTABILITY-019; GitHub Actions run 34979609004, attempt 4; execution head a962ba70077422f8b69642dea4e8272e1d00e5ff; workflow [github/workflows/q041-planck-portability-v19.yml](#); final artifact q041-final-v19, artifact ID 10754769330, SHA-256

BUBBLEVERSE MAIN BOOK

0.42

2284518ecbaaf1ff0d0093977827d5162ec29549ba2b81880f266c250311ca2b. Final state:

CONTROLLED_NO_SCIENTIFIC_RESULT / MAX_SAMPLES_WITHOUT_CONVERGENCE.]

[Internal computational provenance: R-Q040-EDE-CMBSPACE-COUPLED-BRIDGE-005; R-

Q040-MATH-DEFENSIVE-MARGINAL-001; R-Q040-EDE-DEFENSIVE-RQMC-CMB-

MARGINAL-004; PROGRAM_ID Q040-RQMC-V4; GitHub Actions run 34258155387; execution

head 855a28c58246dbdd52d01cae1f40e0611103d96b; workflow .github/workflows/q040-

defensive-rqmc-v4.yml; integration artifact q040-rqmc-integration-final-v4, SHA-256

1cdb34576665d4b91300243ea72b92eab8ee8a24a122fbf118fcd424d891ffc5; final artifact

q040-rqmc-final-v4, SHA-256

450c7a4116eff544d5c8e0f8b209a6baf41caa4cb8b89284923d1956b348a62. Earlier RQMC

V1–V3 executions are technical failures involving serialization, artifact transport, and

deployment and carry no physical evidentiary weight.]

What the Computational Programme Did Not Establish

The computational programme did not establish a mathematical proof of the global minimum of every tested high- H_0 profile.

It did not establish that Planck, SPT-3G, or ACT is incorrect.

It did not establish a faulty Planck calibration.

It did not identify a defective frequency channel.

It did not demonstrate an unmodelled foreground systematic.

It did not establish a covariance defect.

It did not demonstrate that optimizer basins correspond to physically distinct cosmologies.

It did not establish that the historical and later stable-basin labels are directly interchangeable, because the historical continuation and the later portability and revalidation programme used different operational classifier definitions. Nor did the matched primordial-freeze isolation establish that primordial parameterization is globally irrelevant to the historical-versus-later discrepancy; it established only that freezing the primordial block does not recover repeated stable basins under the later common-geometry classifier.

It did not turn repeated internal analyses of the same observational likelihoods into independent observational evidence.

It did not establish whether the CamSpec–HiLLiPoP fitted-geometry difference produces a material downstream cosmological difference, collapses to effectively equivalent constraints, or is conditional on the external dataset combination.

The completed V19 matrix also did not execute the broader planned Λ CDM-versus-EDE comparison with a common supernova block, so its controlled no-science outcome does not close that broader scientific question.

It did not demonstrate new physics.

Q042 - Qualification Closure

After the controlled Q041 no-science endpoint, a successor investigation tested whether the broader downstream-portability contract could be qualified with a new, preregistered and convergence-robust strategy. The frozen scientific contract retained two native Planck likelihood arms, both LCDM and $n=3$ early dark energy, ACT DR6 primary CMB, ACT DR6 lensing, DESI DR2, one identical frozen supernova likelihood, a FULL combination plus four leave-one-out combinations, and the original materiality and model-preference rules. This corresponds to 20 required inference cells.

Q042 is now closed with a documented inconclusive feasibility answer. The strategies actually tested did not establish execution and scientific qualification of that original contract. This is not an affirmative production result and it is not an impossibility theorem: feasibility of a materially different contract-preserving strategy remains unresolved.

The final V31 branch produced partial conditional histories rather than a complete physical reference. Both bounded attempts stopped at the parent 600-second deadline, and no complete tau value was computed. Under the frozen source extension, global rails and append-only retention of the existing certified intervals, however, a separate necessary output-policy obstruction was established. The retained support-candidate counts were 138 for the upper trial and 3,002 for the lower trial against a frozen cap of 32. The certified missing-tail source lower bounds remained above the existing minimum upper bounds, so completing that unchanged retained-interval representation cannot reduce the candidate set enough to satisfy cap32.

That conclusion is deliberately narrow. It rejects more runtime as a sufficient repair of the unchanged V31 stored-interval policy. It does not establish 138 or 3,002 physical minima, an executed tau failure, failure of every alternative numerical method, or a cosmological result. Original-binary arithmetic accuracy, upstream physical and table accuracy, switched-RHS solution properties, a downstream error budget, a valid reference truth, and the convergence-qualified 20-cell comparison all remain unresolved.

The physical universe model therefore does not materially change. The CamSpec-HiLLiPoP fitted-geometry discrepancy remains a methodological tension with unknown downstream significance. $n=3$ early dark energy remains active but constrained; the Hubble-tension benchmarks are unchanged. No production restart is authorized from the Q042 evidence alone.

The immediate next task is operational rather than scientific: synchronize the completed Q042 closure into the repository, registry and candidate model state while preserving the previous accepted snapshot until the update passes its normal gates. Earlier planning language that treated Q042 as an active unchanged execution campaign is superseded by this closure.

Scientific update | Q044 | 9 October 2026

Numerical precision is not yet a cosmological result

An investigation of a frozen optical-depth integration component has established conditional mathematical bounds on a specified spline-based functional. This is a useful numerical qualification, but it is not a new measurement of the expansion rate, a validated correction to the cosmic microwave background, or a comparison of cosmological models.

The relevant calculation distinguishes the native implementation of an optical-depth functional from an ordinary cubic integral. Whether that difference changes the physical reionization history, predicted CMB temperature and polarization spectra, or the likelihood of a cosmological model remains unmeasured in the qualified original analysis.

Therefore the existing Hubble-constant benchmarks and the cautious assessment of early dark energy remain unchanged. Neither equivalence between the CamSpec and HiLLiPoP inference pipelines nor a decisive difference between them has been established by this result.

What remains to be tested

The next proposed investigation isolates the optical-depth integration convention: for the frozen Q041 Lambda-CDM and $n_{\text{scf}} = 3$ early-dark-energy configurations, it asks whether a convergent Thomson-integral reference changes TT, TE and EE predictions and native likelihood values enough to matter for the original H0 and model-comparison decision thresholds.

This is a proposed question, not a completed test or an accepted new model state. Numerical validation, physical prediction accuracy, and posterior convergence remain distinct requirements. Detailed assumptions and provenance are recorded in the technical supplement and the Q044 research journal.

Editorial status and evidence

This section is a dated scientific update to the original edition. It does not silently revise the original chapter pagination or claim that all historical source material has been reprocessed. Primary contextual references: Planck Collaboration, R. Adam et al., Planck intermediate results XLVII, A&A 596 A108 (2016), doi:10.1051/0004-6361/201628897; Lesgourgues (2011), arXiv:1104.2932; Blas, Lesgourgues and Tram (2011), arXiv:1104.2933. Bubbleverse internal evidence: R-Q044-SPLINE-ADJOINT-001 and R-Q044-SPLINE-INGESTION-002.

BUBBLEVERSE MAIN BOOK

0.42

Glossary

AIC — Akaike information criterion. A model-comparison statistic that balances goodness of fit against the number of fitted parameters. A lower AIC can favor one tested model over another, but it is not proof that the model is physically correct.

APlanck — A Planck calibration parameter used in several likelihood analyses in this book. It is a fitted nuisance or calibration quantity, not a separate cosmological observable.

Accepted model — The Bubbleverse-model state that has passed the repository's required promotion and validation process. Candidate content is not accepted merely because it exists.

ACT — Atacama Cosmology Telescope. A ground-based experiment that measures cosmic microwave background temperature and polarization structure at high angular resolution.

Angular power spectrum — A statistical description of how fluctuations on a sky map are distributed across angular scales. CMB temperature and polarization measurements are commonly expressed through TT, TE and EE power spectra.

BAO — Baryon acoustic oscillations. A preferred clustering scale measured statistically in the distribution of matter. Using it as an absolute standard ruler requires calibration of its physical scale.

BBN — Big Bang nucleosynthesis. The early-Universe nuclear process used to predict primordial light-element abundances within a specified physical model.

Best-observed minimum — The lowest likelihood or objective value actually found in the available numerical search. It is not automatically a proven global minimum.

Bubbleverse — The observation-first research framework used in this book to separate measured signals, reconstruction, inference and physical interpretation while preserving a traceable question-by-question research record.

Bubbleverse-model — The dedicated version-controlled repository that stores accepted and candidate Bubbleverse model states, formal model files, tests, provenance and version history.

Calibration — The process of relating an instrument or astrophysical indicator to a known physical scale. Calibration uncertainty can propagate into inferred cosmological parameters.

CamSpec — A Planck CMB likelihood implementation used in the Planck analyses examined in this book.

Candidate model — A proposed next Bubbleverse-model state undergoing validation. A candidate may pass, fail, remain blocked, be revised or be superseded.

CASE — A Bubbleverse provenance identifier used to group a scientific problem or investigation. CASE identifiers are primarily bookkeeping and traceability devices.

BUBBLEVERSE MAIN BOOK

0.42

CMB — Cosmic microwave background. The nearly isotropic microwave radiation field observed across the sky, including small temperature and polarization fluctuations.

Common backend — A deliberately shared computational framework used to compare different models under a controlled set of likelihoods, datasets, priors and numerical conventions. It should not be confused with an exact reproduction of every external paper’s native pipeline.

Confidence interval — A statistical interval constructed according to a specified procedure to express uncertainty in an estimated quantity. Its interpretation depends on the method used to construct it.

Covariance — A description of how uncertainties or residuals vary together. Correlated data cannot generally be treated as a collection of independent measurements.

Dark energy — The name given to the component, field, modification or effective phenomenon used to reproduce accelerated late-time expansion in a cosmological model. The name does not imply that a microscopic dark-energy substance has been directly detected.

Dark matter — The standard description of the additional non-luminous gravitating component required across a wide range of astrophysical and cosmological evidence under conventional gravity. Its microscopic identity remains unknown.

DESI — Dark Energy Spectroscopic Instrument. A spectroscopic survey used in this book for large-scale-structure and baryon-acoustic-oscillation constraints.

Distance ladder — A chain of calibrated distance measurements in which nearby geometric anchors calibrate more distant indicators, eventually allowing the late-Universe expansion rate to be inferred.

EDE — Early dark energy. A class of models in which an additional energy component contributes during an early cosmic epoch and may alter the sound horizon and inferred expansion parameters.

EE — The CMB E-mode polarization auto-spectrum.

Evidence chain — The sequence linking an observed signal through calibration, modelling and statistical inference to a scientific conclusion.

fEDE — The fractional early-dark-energy contribution used in the EDE analyses in this book.

Foreground — Astrophysical emission or contamination that must be modelled or separated from the cosmological signal of interest.

Global minimum — The lowest possible value of an objective function over the complete permitted parameter space. Numerical optimization may find a low minimum without proving that it is the global minimum.

GitHub Actions — GitHub’s automated workflow system. Bubbleverse uses Actions to execute and preserve many numerical programmes and validation pipelines.

BUBBLEVERSE MAIN BOOK

0.42

H_0 — The Hubble constant, the present value of the cosmic expansion parameter $H(t)$. It is inferred from observations through a physical and statistical chain rather than read directly from an instrument.

$H(z)$ — The expansion parameter evaluated at redshift z . Different observational methods constrain it with different calibration and modelling dependencies.

HiLLiPoP — A Planck PR4/NPIPE likelihood implementation independently compared with CamSpec in the later Bubbleverse computational programme.

Hubble tension — The statistically significant disagreement among leading inference chains for the present expansion parameter H_0 . The existence of the discrepancy does not by itself establish its cause or prove new physics.

Inference — A conclusion derived from observations through calibration, modelling, statistics or physical assumptions. Bubbleverse deliberately distinguishes inference from the raw signal recorded by an instrument.

Inference chain — The complete route from observed quantities through calibrations and modelling assumptions to an inferred parameter or conclusion.

JAGB — J-region asymptotic giant branch stars. A stellar distance indicator investigated as an alternative calibration route within the late-Universe distance network.

Λ CDM — Lambda cold dark matter. The standard compact cosmological model containing a cosmological constant, cold dark matter and the usual early-Universe framework. Its empirical success does not make every component a directly observed microscopic entity.

Likelihood — A statistical function describing how compatible specified model parameters are with the observed data under a defined statistical model.

Likelihood architecture — The complete construction of a likelihood, including its data vector, covariance, retained information, nuisance parameters, calibration structure, foreground model and statistical implementation.

Marginalization — A statistical procedure that removes nuisance parameters from the quantity of interest by integrating or otherwise accounting for their allowed values.

Megamaser — Intense microwave emission from molecular gas. Disk megamasers can provide geometric galaxy distances and therefore an alternative route to H_0 .

Mikami — Bubbleverse's scientific triage and revision stage used before the Thin Red Line editing pass. In this book it is a provenance term describing where claims were retained, constrained, rejected or sent to the graveyard.

Model dependence — Dependence of an inferred result on the physical or statistical framework used to translate observations into quantities that are not measured directly.

Multipole, ℓ — The conventional index used to describe angular scale in a spherical-harmonic analysis. Larger ℓ generally corresponds to smaller angular scales.

NPIPE — A Planck data-processing product used in several later Planck likelihood analyses discussed in this book.

BUBBLEVERSE MAIN BOOK

0.42

Nuisance parameter — A fitted parameter required to describe calibration, foregrounds or other non-target effects but which is not itself the main scientific quantity being inferred.

Observation — A signal or quantity recorded from the physical world. Bubbleverse uses the term more narrowly than inference or physical interpretation.

Optimizer basin — A region of parameter space toward which a numerical optimizer can converge. Different starting points may reach different basins.

Pantheon+ — A large Type Ia supernova compilation used in late-Universe distance and cosmological analyses discussed in this book.

Parameter degeneracy — A situation in which different combinations of model parameters can produce similar observational predictions, making the parameters difficult to constrain separately.

Peculiar velocity — Motion of an object relative to the idealized large-scale cosmological expansion. Peculiar velocities matter particularly for low-redshift distance and H_0 inference.

Planck — The ESA space mission that measured the CMB temperature and polarization sky with high precision.

Precision matrix — The inverse of a covariance matrix. It appears in covariance-aware likelihood and residual calculations.

Prior — Information or constraints placed on parameters before or alongside the likelihood in a Bayesian analysis.

Profile likelihood — A likelihood construction in which parameters not currently being examined are reoptimized at each tested value of the parameter of interest.

Provenance — The traceable record connecting a scientific claim or numerical result to its source data, programme, configuration, workflow, commit, artifact or journal entry.

Q — A numbered Bubbleverse research question. Q numbers preserve chronological research provenance; they are not the chapter structure of the main book.

Q firewall — The Bubbleverse-model rule that scientific model state may use only evidence inside the currently authorized Q range.

Q-Journal — The chronological record of a Bubbleverse research question, including its starting state, hypotheses, methods, results, failures, model changes, provenance and handoff.

Redshift, z — The fractional shift of a spectral wavelength between emission and observation. Cosmological redshift is directly observed through spectra; its conversion into distances, ages or an expansion history requires additional modelling.

Reproducibility — The degree to which a calculation can be reconstructed from its data, code, environment, configuration, workflow and recorded provenance.

BUBBLEVERSE MAIN BOOK

0.42

Result ID — A stable Bubbleverse identifier attached to a specific computational or scientific result.

R_d — The sound horizon associated with the baryon-drag epoch. It acts as the physical ruler used in standard interpretations of BAO.

RQMC — Randomized quasi-Monte Carlo. A numerical integration method used in the common CMB-space nuisance-marginalization test without relying on the single-Gaussian approximation. The tested finite implementation did not pass the required numerical-stability criteria.

Scale factor, $a(t)$ — The quantity used in homogeneous and isotropic cosmology to describe the relative cosmic scale as a function of time.

SHOES — The distance-ladder programme using geometric anchors, Cepheid variables, Type Ia supernovae and Hubble-flow objects to infer H_0 .

Sound horizon — The characteristic distance travelled by acoustic disturbances in the early photon-baryon plasma before the relevant decoupling epoch. Its calibrated size is central to BAO and early-Universe inference.

SPT-3G — The third-generation South Pole Telescope camera and its CMB measurements.

Standard candle — An object whose luminosity can be standardized or otherwise calibrated so that its observed brightness provides distance information.

Standard ruler — A physical scale whose calibrated size allows observed angular or redshift separations to provide cosmological distance information.

Standard siren — A gravitational-wave source whose waveform provides luminosity-distance information. Combined with redshift information, it can constrain H_0 .

Superseded result — A historical result retained for provenance after a later analysis provides a corrected or more authoritative result. Superseded does not mean that the historical calculation never occurred.

TE — The cross-spectrum between CMB temperature fluctuations and E-mode polarization.

Thin Red Line — The Bubbleverse editorial boundary that permits restructuring, compression and clarification while forbidding unauthorized changes to the surviving scientific state.

TRGB — Tip of the red-giant branch. A stellar luminosity feature used as a distance indicator and alternative calibration route.

TT — The CMB temperature auto-spectrum.

Uncertainty — A quantitative description of the range or distribution of values consistent with an analysis. Statistical, systematic and other uncertainty components must not be silently merged when the source keeps them separate.

World Breaker / Hulk Smash — Bubbleverse falsification and contradiction-testing stages used before the final structural editing pass. They belong to the research workflow rather than to the cosmological model itself.

BUBBLEVERSE MAIN BOOK

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Research qualification and the current evidence boundary

The cosmological interpretation in this edition remains bounded by the evidence available through the Q042 investigation. The Hubble tension and the question of whether the fitted geometry of $n=3$ early dark energy is portable across Planck likelihood implementations remain unresolved.

The subsequent integration investigation established the internal consistency of a proposed repository update, not a new cosmological measurement. A locally validated candidate is not an accepted model release. The scientific interpretation therefore remains unchanged, and no inference of early-dark-energy detection, falsification, or resolution of the Hubble tension follows from that integration work.

Scientific qualification after the optical-depth investigation

Status: Q045 CLOSED / PHYSICAL RESULT INCONCLUSIVE. Proposed Q046 not yet registered.

The evidence boundary

The current investigation distinguishes a verified numerical recovery from a validated physical prediction. The recovered electron-history diagnostics and integration checks are reproducible technical results, but they do not establish that the full Thomson optical depth, the CMB temperature and polarization spectra, or cosmological likelihood values are accurate enough to change model comparisons.

What remains unresolved

The optical-depth reference-truth gate remains UNQUALIFIED and the final physical-materiality result remains UNRESOLVED. No new observation, Hubble-constant estimate, early-dark-energy preference, or likelihood-portability classification follows. The discrepancy between local and early-universe H_0 inference chains remains open, as does the Planck likelihood-implementation tension.

The next scientific test

A proposed next investigation, Q046 (pending canonical registration), would compare the frozen electron-fraction and cumulative Thomson-depth histories with an independently integrated, event-resolved solution of the same physical model. Each of the four recovered Q041 configurations would hold its accepted L0 reionization redshift fixed. Agreement or disagreement would have to be judged against a justified, preregistered error envelope. This work has not yet supplied a qualified reference.